

Efficient Untying of Weight-Tied Language Models via Asymmetric Geometric Decoupling

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Abstract

Weight tying reduces vocabulary-layer parameters, but forces input embeddings and output unembeddings to share one vocabulary geometry. Modern language models increasingly untie these matrices at larger scales, raising the question of whether parameter-constrained models can relax tying without paying the full cost of an independent vocabulary matrix. We diagnose this trade-off through training dynamics and embedding geometry across public model families. Tied shared matrices are consistently closer to untied output unembeddings than to input embeddings; output spaces also show stronger cross-model alignment and stronger mean/rank-1 spectral structure. These patterns match dense softmax gradients on the output side versus sparse lookup gradients on the input side, suggesting that hard tying makes the shared matrix output-dominated and compresses input-side geometry. We propose **Asymmetric Geometric Decoupling (AGD)**, which keeps a shared vocabulary matrix on the output side while adding small structured freedom to the input side through bias correction and hidden-dimensional low-rank deformation. In MiniMind and FineWeb-Edu/GPT-2 pretraining, input-side AGD variants consistently improve over the tied baseline with far lower overhead than full untying, while output-side counterparts are less stable.

1 Introduction

Weight Tying (WT), sharing the input embedding matrix and the output unembedding matrix, is a classic parameter-efficiency design in neural language models. Press and Wolf (Press and Wolf, 2017) and Inan et al. (Inan et al., 2017) show that output classifier weights themselves can serve as effective word representations, allowing input embeddings and output unembeddings to share the same vocabulary matrix while reducing parameters and preserving or even improving language modeling performance. This design also embeds a strong

structural premise: token semantic encoding on the input side and vocabulary prediction on the output side can be served by the same lexical geometry.

This classic design was inherited in the modern LLM era, but it did not become a universal default. Recent public models show a scale-dependent split: smaller models often keep tied designs to save vocabulary-layer parameters, whereas larger or MoE models more often adopt untied designs to gain greater representational freedom (full inventory in Section B.1). Table 1 illustrates why this split is natural: in the Qwen3 dense series, a single vocabulary matrix accounts for 26.10% of Qwen3-0.6B but only 2.37% of Qwen3-32B. Full untying is therefore expensive for small models but increasingly affordable at larger scales.

This trend raises the central question of this paper: **can parameter-constrained language models relax the geometry imposed by weight tying without paying the full cost of an independent vocabulary matrix?** We argue that the cost of hard tying is not merely one fewer matrix, but the assumption that input embeddings and output unembeddings should share the same lexical geometry.

Our diagnostics show that this assumption is asymmetric in practice. Across public model families, tied shared matrices are more output-like than input-like: they are closer to untied output unembeddings, output spaces show stronger cross-model alignment, and output/tied matrices have stronger mean/rank-1 spectral structure. This geometry is consistent with the underlying training dynamics. Input embeddings receive sparse lookup-path gradients, whereas output unembeddings receive dense full-vocabulary softmax gradients; after tying, both signals update the same matrix, making the shared vocabulary space more easily dominated by output-side objectives.

Based on this diagnosis, we propose **Asymmetric Geometric Decoupling (AGD)**, a family of lightweight embedding-head untying operations

Table 1: **Parameter share of a single embedding matrix in Qwen3 dense models.** A single Emb denotes $V \times d$. For untied models, input embeddings and output unembeddings each have a matrix of the same size, so the “actual total vocabulary-matrix share” is twice the single-Emb share.

Model	tied?	d	Single Emb Params	Total Params	Single Emb / Total	Actual Vocab Matrices / Total
Qwen3-0.6B	tied	1024	155.6M	596.0M	26.10%	26.10%
Qwen3-1.7B	tied	2048	311.2M	1.721B	18.08%	18.08%
Qwen3-4B	tied	2560	389.0M	4.022B	9.67%	9.67%
Qwen3-8B	untied	4096	622.3M	8.191B	7.60%	15.20%
Qwen3-14B	untied	5120	777.9M	14.768B	5.27%	10.53%
Qwen3-32B	untied	5120	777.9M	32.762B	2.37%	4.75%

between hard tying and full untying. AGD keeps a shared vocabulary matrix for the output side while adding small structured freedom to the input side through bias correction and hidden-dimensional low-rank deformation. In MiniMind and FineWeb-Edu/GPT-2 pretraining, input-side AGD variants consistently improve over the tied baseline with far lower overhead than full untying, while output-side counterparts are less stable.

Our main contributions are summarized as follows:

(1) Diagnosis of hard tying. We provide cross-model geometric and spectral evidence that tied vocabulary matrices are output-like rather than neutral compromises between input and output roles.

(2) Mechanistic explanation. We connect this output-like geometry to dense-output versus sparse-input gradient asymmetry, explaining why input-side geometry is the side most likely to be compressed.

(3) Efficient untying. We introduce input-side-prioritized AGD and validate that bias correction and A-LORA improve tied small-model pretraining with minimal parameter overhead.

2 Related Work

Weight tying and its implicit assumption

Weight tying was originally introduced to reduce vocabulary-layer redundancy between input embeddings and output classifiers in neural language

models. Press and Wolf (Press and Wolf, 2017) show that output unembedding weights themselves can serve as effective word representations and observe that tied shared matrices evolve closer to output-side word representations in untied models; Inan et al. (Inan et al., 2017) explain the rationale for sharing input/output vocabulary matrices from the loss-function and word-embedding-space classification perspectives. We revisit this assumption: in modern language models, is a shared matrix still suitable for both input semantic encoding and output vocabulary prediction?

Geometric constraints on input/output vocabulary spaces

Recent work has begun to examine the geometric compatibility of input embeddings and output unembeddings. Bertolotti and Cazzola (Bertolotti and Cazzola, 2024) argue that input embeddings tend to encode semantic similarity, whereas output-side word representations are shaped more by contextual prediction distributions; hard tying therefore imposes nontrivial geometric constraints. Machina and Mercer (Machina and Mercer, 2024) observe from an isotropy perspective that some highly isotropic large models use untied embedding/unembedding weights, but this remains mainly a correlational observation. By contrast, we provide cross-model GCorr and static spectral diagnostics and further propose a low-cost untying mechanism.

Table 2: Models used in the empirical analysis.

MODEL FAMILY	COVERAGE (PARAMS)	ARCHITECTURE	COUNT
QWEN-2.5	0.5–72B	DENSE	14
QWEN-3/3.5/3.6	0.6B–397B	DENSE / MoE	28
LLAMA-3.X	1–70B	DENSE	8
GEMMA-2/3	1–27B	DENSE	12
DEEPSEEK	V2/V3/R1	DENSE / MoE	16
GLM	GLM-5/5.1	MoE	2
MINIMAX	M2.5/M2.7	MoE	2
TOTAL	0.5B–754B	DENSE / MoE	82

Relaxing hard tying and embedding/head decoupling

Prior work has attempted to relax hard tying through structured output layers, intermediate prediction spaces, or capacity reallocation. Pappas et al. (Pappas et al., 2018) model structured sharing between input and output embeddings in neural machine translation; Gulordava et al. (Gulordava et al., 2018) decouple hidden-state space from word-embedding prediction space while preserving tying; REMBERT (Chung et al., 2021) reallocates capacity between input embeddings and output vocabulary representations through re-embedding. These methods show that vocabulary entry and exit need not be strictly identical, but they usually require additional mappings, output-layer changes, or capacity redistribution. AGD has a different goal: it preserves the tied vocabulary matrix as a parameter-efficient base and adds only a small amount of structured freedom on the input side.

Training dynamics and cross-model geometry

Training dynamics also support input/output asymmetry. Press and Wolf (Press and Wolf, 2017) already note that input embeddings receive sparse updates only when tokens appear, whereas output unembeddings receive dense full-vocabulary updates at every step through softmax. We connect this dense-output/sparse-input imbalance to geometric diagnostics in modern models, explaining why tied matrices exhibit output-like structure. On the other hand, cross-model studies such as Lee et al. (Lee et al., 2025) find shared geometric structure in unembedding spaces, supporting our use of embedding/unembedding geometry as the central object for revisiting weight tying.

3 Diagnostic Analysis: The Geometry of Weight Tying

3.1 Experimental Setup

We construct a diagnostic set from major public language-model families, covering dense and MoE architectures, tied and untied designs, and model sizes from 0.5B to 754B parameters (full inventory in Table 2). The analysis consists of 35 Base-Instruct comparisons, 110 within-family model-series comparisons including 37 tied-untied pairs, and static vocabulary-matrix spectral analysis over 82 comparable public models.

Diagnostic protocol Let the input embedding matrix be $E \in \mathbb{R}^{V \times d}$ and the output unembedding matrix be $U \in \mathbb{R}^{V \times d}$; under tied parameter-

ization, both share the same matrix W . **GCorr** (Global Correlation) measures agreement between token-token geometries induced by two aligned vocabulary matrices: for $X, Y \in \mathbb{R}^{n \times d}$ and a pairwise metric μ , it is the Pearson correlation between $[\mu(X_{i_k}, X_{j_k})]_k$ and $[\mu(Y_{i_k}, Y_{j_k})]_k$ on shared token pairs (formal definition and sampling protocol in Section A.3). We report cosine, Euclidean, and squared-Euclidean variants in the main diagnostic comparisons, and complement GCorr with static spectral analysis of the mean axis, the leading singular component, and raw rank-1 energy. Token alignment, bootstrap, and spectral subset construction are given in Sections A.1 and B.2.

3.2 Geometric Differences and the Conflict of Weight Tying

We summarize the diagnostics into three points: tied shared matrices are closer to output unembeddings; $U-U$ GCorr is systematically higher than $E-E$ GCorr in the same cross-model comparisons; and both output unembeddings and tied shared matrices exhibit a stronger mean/rank-1 spectral axis.

Insight 1: tied shared matrices are closer to output unembeddings

In within-family tied-untied comparisons, geometric similarity on the input and output sides shows strong asymmetry. Averaged over 37 tied-untied pairs, output-side similarity is substantially higher than input-side similarity: $U_{\text{cos}} = 0.8114$, $U_{\text{euc}} = 0.8169$, versus $E_{\text{cos}} = 0.1395$, $E_{\text{euc}} = 0.5553$. Pairwise inspection shows that all 37/37 pairs satisfy both $U_{\text{cos}} > E_{\text{cos}}$ and $U_{\text{euc}} > E_{\text{euc}}$. This indicates that when a tied model is compared with an untied model from the same family, the tied shared matrix is closer to the output unembedding in the untied model than to its input embedding.

Static spectrum results provide an independent line of evidence. In the comparable tied subset, 34 models show U-like spectra rather than E-like spectra: the tied median raw rank-1 energy is 11.4%, compared with 5.3% for untied U and 1.0% for untied E; the tied median $\|\mu\|/\text{mean}\|x_i\|$ is 0.32, compared with 0.21 for untied U and 0.09 for untied E. Cross-model GCorr and static spectra jointly indicate that shared tied matrices inherit U-side properties; as a supplement, 13 untied and 17 tied Base-Instruct pairs show that untied input embeddings have essentially unchanged E_{euc} (13/13 ≥ 0.999), whereas tied shared matrices keep high $E_{\text{cos}} \approx 0.99$ but exhibit clear E_{euc} scale drift (see

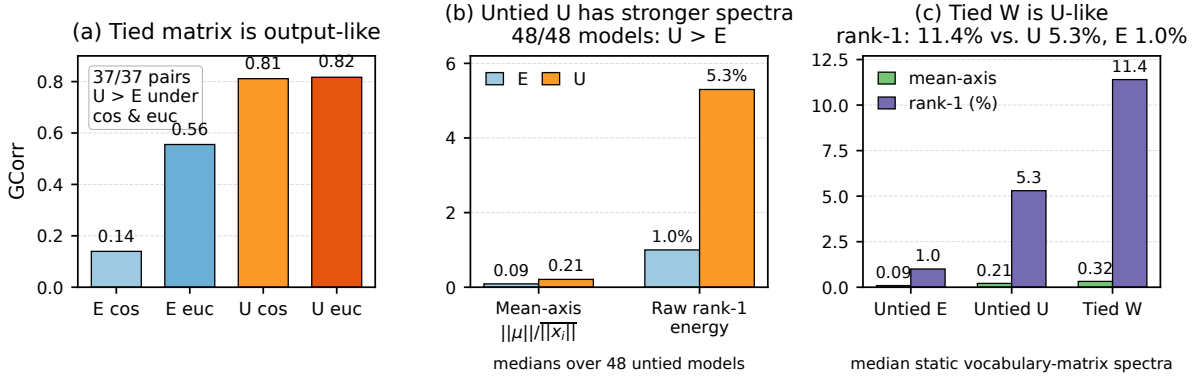


Figure 1: **Three core pieces of evidence from the updated weight-tying geometry diagnostic.** (a) In 37 within-family tied–untied GCorr comparisons, the tied shared matrix is more similar to the untied output unembedding than to the untied input embedding in all 37/37 pairs under both cosine and Euclidean metrics. (b) Among 48 comparable untied models, the U-side static spectrum has a much stronger mean/rank-1 axis than the E side: the median $\|\mu\|/\text{mean}\|x_i\|$ is about $2.3\times$ larger, and raw rank-1 energy is about $5.2\times$ larger. (c) For 34 comparable tied models, the shared matrix falls in the U-like spectral regime: median raw rank-1 energy is 11.4%, compared with 5.3% for untied U and 1.0% for untied E.

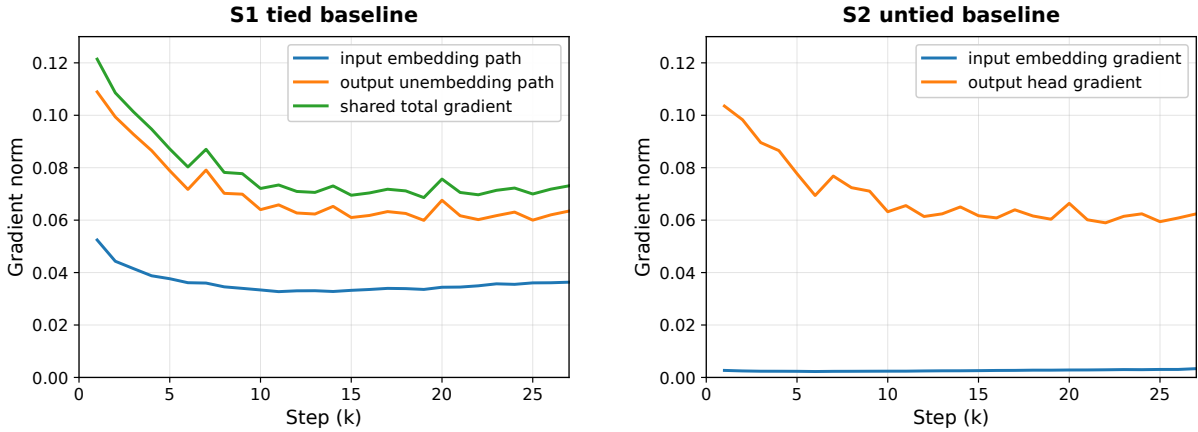


Figure 2: **Input-side and output-side gradient norms in FineWeb-Edu/GPT-2 pretraining.** Curves are averaged over three seeds. In the S1 tied baseline, gradients on the shared matrix are largely elevated by the output-side path; in the S2 untied baseline, input embedding gradients are much smaller than output unembedding gradients. This shows that dense-output/sparse-input update asymmetry pushes tied matrices toward output-side geometry.

Section B.4.1).

Gradient-level observations provide a mechanistic explanation for the geometric results above. Output unembeddings receive dense full-vocabulary updates at every position through the softmax classification objective, whereas input embeddings receive sparse updates only through the lookup path of currently observed tokens; under tying, the two gradient paths are merged onto the same W , so the shared matrix serves the input role while evolving long-term toward output unembedding objectives. The full dense–sparse derivation is given in Section A.2.

Insight 2: U – U GCorr is higher than E – E GCorr Output-side dominance is not limited to tied–untied comparisons. In the updated within-family model-series diagnostics, we compare input-side and output-side geometries over the same cross-model pairs and find substantially higher U – U GCorr than E – E GCorr, especially under cosine geometry. Across the 110 Task2 cross-model pairs, $U_{\text{cos}} > E_{\text{cos}}$ in 85/110 cases (mean U_{cos} 0.83 vs. E_{cos} 0.48; see Sections B.4 and B.4.2). This means that the output vocabulary classifier space is not merely what tied matrices are pulled toward; it also exhibits a more shared geometry across comparable model-series comparisons.

This conclusion is intentionally stated as a cross-

model trend rather than an absolute claim for every metric and every model family. Among the 25 pairs with $U_{\cos} \leq E_{\cos}$, 24 are tied-side ties ($E = U$) or DeepSeek same-family numerical ties at $E_{\cos} \approx 1$; only MiniMax-M2.5-Chat vs. MiniMax-M2.7 is a clear exception (Table 25). Euclidean GCorr can be affected by family-specific scaling choices, whereas cosine GCorr more consistently shows that U spaces align across scales and related architectures. The important point for weight tying is that the output side already has a strong common geometry in untied models, making it a plausible attractor when E and U are forced to share one matrix.

Insight 3: output unembeddings have a stronger mean/rank-1 spectral axis The updated static spectral analysis further clarifies what distinguishes U from E . Among 48 comparable untied models, the U-side raw spectrum is more concentrated almost entirely in the leading component: the median $\|\mu\|/\text{mean}\|x_i\|$ is 0.21 for U versus 0.09 for E (about $2.3\times$), and the median raw rank-1 energy fraction is 5.3% for U versus 1.0% for E (about $5.2\times$), with U rank-1 energy exceeding E in all 48/48 cases.

Crucially, this is not broad low-rank concentration across the first several principal directions. Incremental rank2–5 and rank6–10 energy are lower on U than on E in all 48/48 comparable untied models. The U-side difference is therefore best described as a *mean-axis / rank-1 effect*: the vocabulary classifier matrix carries a stronger global mean component, and its leading singular direction is almost aligned with this mean direction. After centering, the U-side rank-1 advantage remains but becomes much smaller.

Tied models follow the same pattern. In the 34-model comparable tied subset, the shared matrix has median raw rank-1 energy of 11.4% and median $\|\mu\|/\text{mean}\|x_i\|$ of 0.32, both closer to or stronger than the untied U profile and far from the diffuse untied E profile. This spectral evidence reinforces the GCorr results: tying does not produce an intermediate neutral vocabulary space; it gives the shared matrix U-like mean-axis structure.

Taken together, these three points show that standard weight tying is not a symmetric compromise between input embeddings and output unembeddings: the shared matrix is geometrically closer to output unembeddings, U – U GCorr is systematically higher than E – E GCorr in comparable cross-

model comparisons, and both U and tied matrices exhibit a stronger mean/rank-1 spectral axis than E. The next step is therefore to preserve tied parameter efficiency while giving the input side limited structured freedom to move away from the output-dominated shared geometry. This is the starting point of asymmetric geometric decoupling in the next section.

4 Method: Asymmetric Geometric Decoupling

The diagnosis in Section 3.2 shows that standard weight tying is not neutral sharing between the input and output sides: the shared vocabulary matrix is geometrically closer to output unembeddings and inherits their stronger mean/rank-1 structure. Based on this observation, we study **Asymmetric Geometric Decoupling (AGD)**: preserving a shared vocabulary matrix as a parameter-efficient base while adding a small amount of structured freedom only on the input side, allowing input embeddings to deviate from the output-dominated shared space.

4.1 From Hard Tying to Input-Side AGD

Let the vocabulary size be V and the hidden dimension be d . A standard tied language model uses the same matrix $W \in \mathbb{R}^{V \times d}$ for both input embedding and output unembedding:

$$e_t = W[x_t], \quad z_t = Wh_t + b, \quad (1)$$

where h_t is the final-layer hidden state and b is the output bias. Equivalently, standard tying imposes the hard constraint

$$E_{\text{eff}} = U_{\text{eff}} = W. \quad (2)$$

AGD relaxes this hard equality rather than introducing a full new vocabulary matrix. We keep

$$U_{\text{eff}} = W, \quad (3)$$

so the output side continues to use the shared matrix and dense full-vocabulary classification gradients; and we set

$$E_{\text{eff}} = \phi_{\text{in}}(W; \theta), \quad (4)$$

where ϕ_{in} is an input-side transformation with only a small number of parameters θ . Since the tied matrix already behaves output-like, limited extra parameters should be allocated to the compressed input side rather than further strengthening the output side. The full dense-output/sparse-input gradient derivation is given in Section A.2.

Table 3: Four input-side AGD variants used in the main text. All variants keep $U_{\text{eff}} = W$.

Variant	Effective Input Matrix E_{eff}	Added
S3: bias	$W - \mathbf{1}\beta^\top$	d
S4: add. LoRA	$W + A_{\text{in}}B_{\text{in}}$	$r(V + d)$
S6: mult. LoRA	$W(I_d + sP_{\text{in}}Q_{\text{in}})$	$2dr$
S12: A-LoRA	$W(I_d + sP_{\text{in}}Q_{\text{in}}) - \mathbf{1}\beta^\top$	$d + 2dr$

4.2 Four Input-Side Variants in the Main Text

The main text focuses on four input-side AGD variants (S3, S4, S6, and S12); output-side counterparts, the untied extension, and the full S1–S12 definitions are in Section B.3. All four keep $U_{\text{eff}} = W$ and differ only in the input-side ϕ_{in} . Variants with hidden-dimensional multiplicative LoRA (S6/S7/S12) use $W(I_d + sPQ)$, where $s > 0$ scales the low-rank deformation; all main pretraining experiments use $s = 1$.

S3: input-side bias correction S3 learns a hidden-dimensional bias vector $\beta \in \mathbb{R}^d$:

$$E_{\text{eff}} = W - \mathbf{1}\beta^\top, \quad U_{\text{eff}} = W. \quad (5)$$

This form is the minimal response to the mean/rank-1 diagnosis: if W is already biased toward an output-like origin, the input side needs only d parameters to learn a global translation that corrects input embeddings away from that mean axis. Formally, $\mathbf{1}\beta^\top$ is a rank-1 affine component whose vocabulary-side coefficient vector is fixed to all ones, so it does not introduce token-specific memory.

S4: input-side additive LoRA S4 adds a standard low-rank residual on the vocabulary matrix:

$$E_{\text{eff}} = W + A_{\text{in}}B_{\text{in}}, \quad A_{\text{in}} \in \mathbb{R}^{V \times r}, \quad B_{\text{in}} \in \mathbb{R}^{r \times d}. \quad (6)$$

It allows different low-rank coefficients per token and is more expressive than a global bias, but its parameter count $r(V + d)$ grows linearly with vocabulary size. S4 is therefore mainly used to test whether vocabulary-dimensional low-rank freedom is worth its extra cost.

S6: input-side multiplicative LoRA S6 does not add token-specific coefficients in the vocabulary dimension, but instead learns a low-rank transformation in hidden semantic coordinates, with $P_{\text{in}} \in \mathbb{R}^{d \times r}$, $Q_{\text{in}} \in \mathbb{R}^{r \times d}$, and $s > 0$:

$$E_{\text{eff}} = W(I_d + sP_{\text{in}}Q_{\text{in}}). \quad (7)$$

It adds only $2dr$ parameters and does not grow with V ; s adds no trainable matrices and only scales the low-rank correction, with $s = 1$ in all main experiments. Geometrically, S6 applies a shared low-rank deformation along hidden directions to all tokens. A related multiplicative form also appears in LoRMA (Bihany et al., 2025).

S12: A-LoRA S12 combines the global translation of S3 with the hidden-dimensional low-rank deformation of S6. Since this is an input-side affine transformation of the shared vocabulary vectors, we name it A-LoRA (Affine LoRA):

$$E_{\text{eff}} = W(I_d + sP_{\text{in}}Q_{\text{in}}) - \mathbf{1}\beta^\top. \quad (8)$$

Here $A_{\text{aff}} = I_d + sP_{\text{in}}Q_{\text{in}}$ is the low-rank-parameterized linear part and $c = -\beta$ is the translation term, so $E_{\text{eff}} = WA_{\text{aff}} + \mathbf{1}c^\top$ has standard affine form; main experiments again use $s = 1$. With $d + 2dr$ parameters, S12 covers mean-axis correction and low-rank coordinate deformation along a few hidden directions without growing with vocabulary size. Subsequent experiments compare S3/S4/S6/S12 against tied and full untied baselines to separate the roles of diagnosis-aligned minimal bias correction and general low-rank input-side freedom.

5 Experimental Evaluation

We evaluate a set of embedding-head lightweight untying variants with the goal of verifying whether structurally relaxing input embeddings and output unembeddings can consistently alleviate the geometric constraints of weight tying without introducing the cost of a full independent vocabulary matrix. Experiments cover only the pretraining stage and do not include SFT, RL, DPO, or other post-training pipelines.

5.1 Experimental Setup

All variants use the same lightweight dense decoder-only Transformer architecture with hidden dimension $d = 768$ and output bias enabled (lm_head_bias), aligned with the Qwen3 dense model design (Yang et al., 2025). Low-rank variants uniformly use rank $r = 32$. We set up two pretraining data and tokenizer configurations: the first uses MiniMind (Gong, 2025) data and the MiniMind tokenizer, training all twelve embedding-head variants; the second uses FineWeb-Edu (Penedo et al., 2024) data and the GPT-2 tokenizer (Radford et al., 2019), training

Table 4: Three-seed validation results in both pretraining settings. Seed Δ PPL is the same-seed PPL difference from S1; PPL Δ is the mean percentage change relative to S1; PPL Δ /Param Δ follows the “reduction-is-positive” convention. Full ablations are in the appendix.

Data Setting	Variant	Param Δ	Mean Loss	Mean PPL	Seed Δ PPL vs S1	PPL Δ	PPL Δ /Param Δ
MiniMind	S1: tied baseline	0.0000%	1.914962	6.786681	0.000 ± 0.000	0.00%	–
MiniMind	S3: bias	0.0012%	1.903291	6.707934	-0.079 ± 0.040	-1.16%	966.93
MiniMind	S12: A-LORA	0.0781%	1.906849	6.731843	-0.055 ± 0.036	-0.81%	10.35
MiniMind	S4: add. LoRA	0.3589%	1.912134	6.767515	-0.019 ± 0.044	-0.28%	0.79
MiniMind	S6: mult. LoRA	0.0769%	1.912824	6.772186	-0.014 ± 0.031	-0.21%	2.78
MiniMind	S2: full untied baseline	7.6898%	1.977121	7.221921	$+0.435 \pm 0.029$	+6.41%	–
FineWeb-Edu/GPT-2	S1: tied baseline	0.0000%	3.237896	25.480064	0.000 ± 0.000	0.00%	–
FineWeb-Edu/GPT-2	S12: A-LORA	0.0511%	3.230316	25.287638	-0.192 ± 0.071	-0.76%	14.78
FineWeb-Edu/GPT-2	S3: bias	0.0008%	3.233494	25.368130	-0.112 ± 0.082	-0.44%	549.13
FineWeb-Edu/GPT-2	S4: add. LoRA	1.6722%	3.235201	25.411471	-0.069 ± 0.069	-0.27%	0.16
FineWeb-Edu/GPT-2	S6: mult. LoRA	0.0503%	3.235579	25.421078	-0.059 ± 0.080	-0.23%	4.60
FineWeb-Edu/GPT-2	S2: full untied baseline	39.5284%	3.311899	27.437170	$+1.957 \pm 0.060$	+7.68%	–

key input-side and output-side lightweight untying variants on roughly 6 billion tokens of packed corpus. All runs use three random seeds (42, 123, 2026) and report mean validation cross-entropy loss and PPL; detailed data splits, training hyperparameters, parameter overhead, gradient logging, and log paths are given in Section B.3.

To systematically study structured relaxation of tied input embedding/output unembedding matrices and its effect on language model pretraining, we construct a set of embedding-head variants on the same model backbone. The main text focuses on the tied baseline, the full untied baseline, and input-side lightweight untying variants that improve over the tied baseline in both data settings (S3/S12-A-LORA/S4/S6), highlighting the effective input-side-priority path in AGD. Output-side relaxations are summarized as negative results in the main text, while full variant definitions, effective matrix formulas, and parameter overhead are given in Section B.3.

5.2 Main Results

Input-side relaxation consistently outperforms the tied baseline Both data settings show a consistent trend (Table 4): the most stable gains come from input-side corrections. On MiniMind, input-side bias S3 performs best and S12-A-LORA ranks second; in the more general FineWeb-Edu/GPT-2 setting, S12-A-LORA becomes the best configuration and S3 still yields stable gains. By contrast, the full untied baseline is clearly weaker than the tied baseline in both small-model pretraining settings. This is consistent with classic tying work (Press and Wolf, 2017; Inan et al., 2017) and with Leviathan’s small-language-model com-

parison (Batley and Saha, 2026): naïve full untying does not reliably outperform tying in small-scale language models, so adding a full independent vocabulary matrix is not an effective trade-off under parameter constraints.

More importantly, the parameter efficiency in the table aligns closely with the spectral diagnosis above. S3 adds only one hidden-dimensional bias vector, with parameter overhead of just 0.0012% and 0.0008% in the two settings, yet brings stable PPL reductions in both; its PPL Δ /Param Δ reaches 966.93 on MiniMind and 549.13 on FineWeb-Edu/GPT-2, the strongest per-parameter gains among all lightweight variants. This is consistent with the observation in Section 3 that U/tied matrices have a stronger mean/rank-1 axis: if the main geometric deviation of hard tying first appears as an output-like global mean axis, then input-side rank-1 affine correction is the lowest-cost and most direct fix. In other words, the strong parameter efficiency of S3 is not accidental “small-module regularization,” but an experimental validation of the mean-axis mismatch diagnosis above.

A-LORA may bring additional gains S12 (A-LORA) adds multiplicative LoRA on top of S3’s rank-1 bias correction, equivalent to the input-side affine map $W A_{\text{aff}} + 1c^\top$ that corrects mean-axis offset and adjusts a few hidden directions. This combination achieves the lowest mean PPL on FineWeb-Edu/GPT-2 (PPL Δ /Param Δ =14.78). With only two small-model settings, we read this as bias capturing first-order geometric deviation and multiplicative LoRA supplementing finer input-side freedom at modest cost, rather than as a strong transferability claim.

S4 and S6 also provide useful parameter-form comparisons. S4 uses vocabulary-dimensional additive LoRA of the form $W + A_{\text{in}}B_{\text{in}}$, with parameter count growing linearly with V ; under the GPT-2 vocabulary it adds roughly 1.63M parameters, a 1.6722% overhead, but reduces mean PPL by only 0.27%, yielding a per-parameter gain of 0.16. S6 uses the hidden-dimensional multiplicative form $W(I_d + sP_{\text{in}}Q_{\text{in}})$ with $s = 1$ in the main experiments, with parameter count depending only on d and r and not growing with vocabulary size, so its added parameters are close to A-LORA and it also maintains positive gains in both settings. Its parameter efficiency remains reasonable but is lower than S3/A-LORA, which include explicit mean-axis correction. This comparison shows that the bias term aligned with the rank-1 spectral difference is the most critical and efficient component of AGD; multiplicative LoRA provides a parameter-light hidden-space deformation that is most effective as a supplement to bias correction.

Output-side modifications alone yield unstable gains By contrast, adding lightweight freedom only on the output side (S5, S7, S11) fails to replicate the stable gains of input-side methods and is therefore omitted from the main results table; full numbers are in Tables 20 and 21. This negative result is consistent with the diagnosis above: when the tied matrix is already closer to output-side geometry, limited extra freedom is more effective when allocated to the input side.

6 Conclusion

This paper studies whether parameter-constrained language models can relax the geometric constraints of weight tying without learning a fully independent vocabulary matrix. Our diagnostics indicate that tied shared matrices are not neutral compromises between input embeddings and output unembeddings: they are more output-like in geometry and spectrum, consistent with dense-output versus sparse-input update asymmetry. This motivates Asymmetric Geometric Decoupling (AGD), which keeps the shared vocabulary matrix on the output side while giving the input side limited structured freedom through bias correction and hidden-dimensional low-rank deformation.

In small-model pretraining experiments, input-side AGD variants improve over the tied baseline with much lower overhead than full untying, while full untying and output-side modifications are less

reliable in these settings. More broadly, the main opportunity suggested by this work is that, for modern small language models that still use tied embeddings, AGD-like input-side decoupling offers a direct low-cost candidate between hard tying and full untying.

Limitations

Although our diagnostic analysis covers public model families up to roughly 754B parameters and observes consistent GCorr and spectral trends across the updated public-model inventory, full pretraining ablations remain limited to MiniMind dense-scale small causal language models and mainly cover the pretraining stage. Therefore, although we have validated geometric principles and efficiency gains on both MiniMind original data and FineWeb-Edu/GPT-2 settings, the scalability of these specific untying operations to larger foundation models, long-token training, and post-training pipelines such as SFT/RL/DPO still requires empirical validation. Extending the pretraining study to additional low-rank settings, longer token budgets, and finer analysis of gradient allocation between input-side bias and multiplicative LoRA modules also remains for future work.

Ethical Considerations

This work performs diagnostic analysis and pre-training experiments on publicly available pre-trained language models and standard pretraining corpora. It does not involve human subjects or personally sensitive data. To the authors’ knowledge, there are no additional ethical risks requiring disclosure.

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in the main text; Section A.2 gives the full gradient derivation of dense–sparse update asymmetry; Section A.3 explains the metrics used in later geometric diagnostics; and Section A.4 expands the low-rank relaxation assumptions in AGD.

A.1 Notation Definitions in Main-Text Order

Table 5: Main notation used in this paper. Symbols are arranged roughly in the order they first appear in the main technical narrative; when a symbol is overloaded across contexts, the description notes this explicitly.

Symbol	Meaning	Description
V	Vocabulary size	Number of rows in vocabulary matrices; V may differ across model families.
d	Hidden dimension	Number of columns in vocabulary matrices and Transformer hidden-state dimension.
$\mathbf{x} = (x_1, \dots, x_T)$	Token sequence	$x_t \in \{1, \dots, V\}$ denotes the t -th input token.
$E \in \mathbb{R}^{V \times d}$	Input embedding matrix	Maps discrete tokens to hidden-space vectors.
$e_t = E[x_t]$	Input embedding vector	Input-side representation obtained by lookup for token t .
$U \in \mathbb{R}^{V \times d}$	Output unembedding	Projects final hidden states back to vocabulary logits; <code>lm_head</code> in code corresponds to this object.
W	Shared or output vocabulary matrix	In tied settings, usually the shared matrix serving both input and output roles; in some formulas also denotes the output unembedding matrix.
<i>tied / untied</i>	Tied / untied parameterization	Tied means $E = U = W$; untied means input embeddings E and output unembeddings U are independent.
$h_t \in \mathbb{R}^d$	Final hidden state	Hidden vector output by the Transformer backbone at position t .
z_t	Output logits	Usually written as $z_t = Uh_t + b$; if output bias is omitted, $z_t = Uh_t$.
$b \in \mathbb{R}^V$	Output bias	Can absorb unigram frequency or global calibration terms; our geometric analysis focuses mainly on row-vector structure of E and U .
p_t	Next-token distribution	$p_t = \text{softmax}(z_t)$.
y_t	Target token	In autoregressive language modeling, usually $y_t = x_{t+1}$.
$\ell_t, \mathcal{L}$	Token loss and average loss	$\ell_t = -\log p_t[y_t]$, and $\mathcal{L}$ is mean cross-entropy over sequence positions.
∇W_{out}	Output-side gradient	From the full-vocabulary dense classification objective of output unembeddings.
∇W_{in}	Input-side gradient	From the row-sparse input objective of embedding lookup.
$X, Y \in \mathbb{R}^{n \times d}$	Aligned vocabulary matrices	Two matrices used in geometric diagnostics; may correspond to E - E or U - U comparisons.
μ	Token-pair relation metric	May be cosine similarity, Euclidean distance, or squared Euclidean distance.
$\text{GCorr}_\mu(X, Y)$	Global geometric correlation	Compares pairwise geometric relations induced by X and Y on the same token pairs.
$N_{\text{tok}}, N_{\text{pair}}, N_{\text{fit}}$	Sampling and fitting scale	Token sampling cap, pairwise estimate count, and number of tokens used in affine-mapping fitting.
A, c	Affine-mapping parameters	Used in geometric diagnostics to fit $\hat{Y} = XA + \mathbf{1}c^\top$; distinct from A in LoRA.
$M \in \mathbb{R}^{V \times d}$	Generic vocabulary matrix	In the method section, the base input embedding or output unembedding matrix being relaxed.
ΔM	Vocabulary-matrix increment	Low-rank correction matrix learned by LoRA-style relaxation.
r	LoRA rank	Controls low-rank correction freedom; usually $r \ll d$ and $r \ll V$.
A, B	Additive LoRA factors	In $\Delta M = AB$, $A \in \mathbb{R}^{V \times r}$ and $B \in \mathbb{R}^{r \times d}$.
P, Q	Multiplicative LoRA factors	In $M_{\text{eff}} = M(I_d + sPQ)$, $P \in \mathbb{R}^{d \times r}$ and $Q \in \mathbb{R}^{r \times d}$.
s	Multiplicative LoRA scale	Scalar $s > 0$ controlling low-rank deformation strength; adds no matrix parameters. Main pretraining experiments use $s = 1$.
β	Affine correction vector	Learnable vector for global translation correction on input or output side in AGD; can also be viewed as a rank-1 additive LoRA special case with vocabulary-side coefficients fixed to 1.
$E_{\text{eff}}, U_{\text{eff}}$	Effective input / output matrices	Matrices actually used for lookup and output unembedding projection in S1–S12.

A.2 Full Derivation of Dense-Sparse Update Asymmetry

For notational simplicity, let $\mathbf{y}_t \in \{0, 1\}^V$ denote the one-hot vector of target token y_t . From softmax cross-entropy,

$$\frac{\partial \ell_t}{\partial z_t} = p_t - \mathbf{y}_t. \quad (9)$$

Therefore, the gradient of the output unembedding matrix is

$$\frac{\partial \mathcal{L}}{\partial U} = \frac{1}{T-1} \sum_{t=1}^{T-1} (p_t - \mathbf{y}_t) h_t^\top. \quad (10)$$

For the output row vector U_i corresponding to token i , its gradient is

$$\frac{\partial \mathcal{L}}{\partial U_i} = \frac{1}{T-1} \sum_{t=1}^{T-1} (p_t[i] - \mathbb{I}[y_t = i]) h_t. \quad (11)$$

Because softmax probabilities $p_t[i]$ are usually non-zero across all vocabulary positions, the output side produces task gradients for classifier vectors of nearly all V tokens at every training position. This is the full-vocabulary dense update described in the main text.

By contrast, input embeddings are a lookup operation. Let $g_t = \partial \mathcal{L} / \partial e_t$ denote the gradient back-propagated from the Transformer backbone to the input embedding vector; then the gradient of input embedding row E_i is

$$\frac{\partial \mathcal{L}}{\partial E_i} = \sum_{t: x_t = i} g_t. \quad (12)$$

Only when token i appears in the current input sequence does that row receive input-side task gradients; therefore, the input side is row-sparse. Optimizer terms such as weight decay, momentum, or Adam second moments may add scaling or state updates for unseen tokens, but do not change the sparse nature of task gradients themselves.

Under tying, the same matrix W serves both input and output roles, so its task gradient can be written as

$$\nabla W_{\text{tied}} = \nabla W_{\text{out}} + \nabla W_{\text{in}}. \quad (13)$$

Because ∇W_{out} covers the full vocabulary at every step whereas ∇W_{in} covers only rows of currently observed tokens, the long-term optimization dynamics of the shared matrix are naturally more easily dominated by output-side classification objectives. This asymmetry is the training-dynamics basis for the “output-manifold attractor” and input geometric degradation analyses in the main text.

A.3 Geometric Metrics and Diagnostic Indicators

Geometric diagnostics in the main text do not directly compare element-wise values of two token vectors, but compare token-token relations induced by vocabulary matrices. Given two aligned matrices $X, Y \in \mathbb{R}^{n \times d}$, we compute similarity or distance on the same token pairs (i_k, j_k) and compare Pearson correlation of the two relation vectors:

$$\begin{aligned} \text{GCorr}_\mu(X, Y) &= \text{PearsonCorr}(\mathbf{v}_\mu^X, \mathbf{v}_\mu^Y), \\ \mathbf{v}_\mu^X &:= [\mu(X_{i_k}, X_{j_k})]_{k=1}^m, \\ \mathbf{v}_\mu^Y &:= [\mu(Y_{i_k}, Y_{j_k})]_{k=1}^m. \end{aligned} \quad (14)$$

Here μ includes cosine similarity, Euclidean distance, and squared Euclidean distance. The three metrics answer slightly different questions: cosine focuses more on directional structure and is easily affected by global center offset; Euclidean distance preserves relative distance structure of translated point clouds and is better suited for observing whether input embeddings maintain stable semantic constellations; squared Euclidean serves as a robustness check, amplifying larger distance differences and helping exclude accidental conclusions from a single distance form.

GCorr near 1 indicates that two matrices induce highly consistent token-token geometric structure; near 0 indicates weak geometric correlation; below 0 indicates anti-correlated relations on sampled token pairs. Because computing the full $V \times V$ distance matrix is too expensive, the main text uses Monte Carlo token-pair estimation with bootstrap confidence intervals.

Beyond global relation correlation, the main text also uses affine-mapping tests to measure whether two spaces are linearly connected. Given source matrix X and target matrix Y , we fit

$$\hat{Y} = XA + \mathbf{1}c^\top, \quad (15)$$

where $A \in \mathbb{R}^{d \times d}$ is a linear transform and $c \in \mathbb{R}^d$ is a global translation. If R^2 is high and relative reconstruction error is low, the two spaces differ mainly by an affine transform; if GCorr is high but affine mapping is poor, they may preserve partial relational structure without simple linear alignability. We use both indicator types to distinguish “Euclidean rigidity of input space” from “task-driven deformation of output space.”

A.4 Low-Rank Parameterization and the Low-Rank Subspace Hypothesis

The core motivation of original LoRA is low intrinsic rank during model adaptation: when a pre-trained model is adapted to a specific task or domain, full weight updates ΔW often need not occupy the full rank of the original matrix and can be approximated with low-rank decomposition (Hu et al., 2022). Therefore, original LoRA usually freezes base weights and trains only additional low-rank adaptation matrices; these matrices serve parameter-efficient fine-tuning in post-training and can optionally remain mounted or be merged with base weights at deployment.

Our use of LoRA-style low-rank matrices has a different purpose. We do not add temporary task adapters to an existing model, but use low-rank freedom as part of embedding-head architecture itself to alleviate geometric conflict between tied input embeddings and output unembeddings. In other words, our low-rank hypothesis is that the main differences required on the input and output sides relative to the shared vocabulary matrix need not be a full independent $V \times d$ matrix, but can be expressed by “mobility” along a few shared directions. Therefore, these low-rank matrices are not optional post-training attachments functionally, but permanent components of effective input matrix E_{eff} or effective output matrix U_{eff} .

Given base vocabulary matrix $M \in \mathbb{R}^{V \times d}$, standard additive LoRA restricts the full increment matrix ΔM to low-rank decomposition:

$$\begin{aligned} M_{\text{eff}} &= M + \Delta M, \\ \Delta M &= AB, \\ A &\in \mathbb{R}^{V \times r}, \quad B \in \mathbb{R}^{r \times d}. \end{aligned} \quad (16)$$

where $r \ll \min(V, d)$ and the added parameter count is $r(V + d)$, far smaller than a full additional $V \times d$ matrix introduced by complete untying. The actual assumption is not that correction magnitude is small for each token, but that correction vectors for all tokens jointly lie in the same low-dimensional subspace. Let the low-rank coefficient for token i be $a_i \in \mathbb{R}^r$; then its row correction can be written as

$$\begin{aligned} \Delta m_i &= a_i^\top B, \\ \Delta m_i &\in \text{span}(B_1, \dots, B_r) \subseteq \mathbb{R}^d. \end{aligned} \quad (17)$$

Different tokens may have different coefficients a_i , but the directions they can move in are determined by the same set of r global basis directions;

equivalently, the full correction matrix satisfies $\text{rank}(\Delta M) \leq r$.

Therefore, the low-rank subspace hypothesis contains two constraints: first, main geometric differences on the input or output side relative to the shared vocabulary matrix can be explained by a few global directions; second, differences among tokens appear mainly as combination weights on these directions, rather than requiring many independent high-rank token-specific corrections. Full untying allows each token to adjust its input representation or output unembedding weight independently in $\mathbb{R}^d$; LoRA allows all tokens to adjust only within a shared low-rank correction space. Our weight-space LoRA ablations test whether the freedom required for vocabulary-layer input/output decoupling can be replaced by this low-rank subspace hypothesis.

Affine correction vector β can be viewed as an extreme special case of this additive low-rank form. For input-side translation,

$$E_{\text{eff}} = W - \mathbf{1}\beta^\top \quad (18)$$

is equivalent to letting $r = 1$, $A = \mathbf{1} \in \mathbb{R}^{V \times 1}$, and $B = -\beta^\top \in \mathbb{R}^{1 \times d}$ in $\Delta M = AB$. That is, β does not assign different coefficients to different tokens, but forces all vocabulary rows to translate along the same global direction. Its key advantage is fixing the large vocabulary-side coefficient vector to all ones and learning no V -dependent parameters on the vocabulary side, achieving maximal vocabulary-side parameter savings while still capturing the global origin/centroid offset observed in the main text.

Compared with additive LoRA, hidden-dimensional multiplicative LoRA places low-rank freedom in hidden coordinate transforms:

$$\begin{aligned} M_{\text{eff}} &= M(I_d + sPQ), \\ P &\in \mathbb{R}^{d \times r}, \quad Q \in \mathbb{R}^{r \times d}, \quad s > 0. \end{aligned} \quad (19)$$

It introduces only $2dr$ parameters and does not grow with vocabulary size V ; s is an additional scalar hyperparameter that damps or amplifies the low-rank correction, and all main pretraining experiments use $s = 1$. Geometrically, it does not add independent corrections for each token, but learns a low-rank deformation in hidden dimensions, allowing the full input embedding or output unembedding matrix to relax controllably along shared low-dimensional directions. This form directly supports the AGD design in the main text of

“repairing shared geometry with a small amount of structured freedom.”

B Detailed Experimental Setup

This appendix supplements the experimental protocol, hyperparameter configuration, and the complete model comparison list used in the diagnostic analysis.

B.1 Analysis Model Inventory and Vocabulary-Matrix Configuration

The model inventory used for tied/untied status analysis comes from a centrally maintained configuration with 94 public models: 44 tied and 50 untied, all with determinable vocabulary-matrix sharing status. We fetch public source files for these models and extract input embedding and output unembedding matrices for analysis; because matrix shapes match in this inventory, we report vocabulary-matrix dimensions uniformly. Total parameters mainly follow nominal counts from public names or model cards; for R1-Distill models, because DeepSeek release versions untie vocabulary matrices, we report parameter counts converted from actual safetensors weights via Hugging Face metadata. For MoE models, this column reports total parameters rather than per-token activated parameters. The main-text empirical analysis (Table 2; Section 3) uses a comparable subset of 82 models—48 untied and 34 tied—obtained by excluding 12 inventory entries: the Gemma-4 family (8 models), Gemma-3-1B / Gemma-3-1B-Instruct (2 models), and DeepSeek-V4-Flash / Pro (2 models). These models are retained in the inventory below to document public tied/untied configurations, but their static-spectrum or output-head conventions are not directly comparable to the other models; the exclusion rationale is documented in the notes below and in Section B.4.

For readability, the inventory is split into multiple tables by model ecosystem. We analyze only text token embeddings and output unembeddings, and do not include vision encoders, cross-modal projectors, or other non-vocabulary modules in multimodal models.

Table 6: Qwen model inventory and vocabulary-matrix configuration. The Qwen family (Qwen Team, 2025; Yang et al., 2025; Qwen Team, 2026a,b) covers dense, MoE, and some natively multimodal variants, and is one of the main samples for scale-stratified tied/untied phenomena in the main text.

Model Group	Model	Total Params	Status	Vocab Matrix Dim
Qwen-2.5	Qwen2.5-0.5B	0.5B	tied	151936×896
	Qwen2.5-0.5B-Instruct	0.5B	tied	151936×896
	Qwen2.5-1.5B	1.5B	tied	151936×1536
	Qwen2.5-1.5B-Instruct	1.5B	tied	151936×1536
	Qwen2.5-3B	3B	tied	151936×2048
	Qwen2.5-3B-Instruct	3B	tied	151936×2048
	Qwen2.5-7B	7B	untied	152064×3584
	Qwen2.5-7B-Instruct	7B	untied	152064×3584
	Qwen2.5-14B	14B	untied	152064×5120
	Qwen2.5-14B-Instruct	14B	untied	152064×5120
	Qwen2.5-32B	32B	untied	152064×5120
	Qwen2.5-32B-Instruct	32B	untied	152064×5120
	Qwen2.5-72B-Base	72B	untied	152064×8192
	Qwen2.5-72B-Instruct	72B	untied	152064×8192
Qwen-3	Qwen3-0.6B-Base	0.6B	tied	151936×1024
	Qwen3-0.6B	0.6B	tied	151936×1024
	Qwen3-1.7B-Base	1.7B	tied	151936×2048
	Qwen3-1.7B	1.7B	tied	151936×2048
	Qwen3-4B-Base	4B	tied	151936×2560
	Qwen3-4B	4B	tied	151936×2560
	Qwen3-8B-Base	8B	untied	151936×4096
	Qwen3-8B	8B	untied	151936×4096
	Qwen3-14B-Base	14B	untied	151936×5120
	Qwen3-14B	14B	untied	151936×5120
	Qwen3-32B	32B	untied	151936×5120
	Qwen3-30B-A3B-Base	30B	untied	151936×2048
	Qwen3-30B-A3B	30B	untied	151936×2048
	Qwen3-235B-A22B	235B	untied	151936×4096
Qwen-3.6	Qwen3.6-35B-A3B	35B	untied	248320×2048
Qwen-3.5 Dense	Qwen3.5-0.8B-Base	0.8B	tied	248320×1024
	Qwen3.5-0.8B-Instruct	0.8B	tied	248320×1024
	Qwen3.5-2B-Base	2B	tied	248320×2048
	Qwen3.5-2B-Instruct	2B	tied	248320×2048
	Qwen3.5-4B-Base	4B	tied	248320×2560
	Qwen3.5-4B-Instruct	4B	tied	248320×2560
	Qwen3.5-9B-Base	9B	untied	248320×4096
	Qwen3.5-9B-Instruct	9B	untied	248320×4096
	Qwen3.5-27B-Instruct	27B	untied	248320×5120
Qwen-3.5 MoE	Qwen3.5-35B-A3B-Base	35B	untied	248320×2048
	Qwen3.5-35B-A3B-Instruct	35B	untied	248320×2048
	Qwen3.5-122B-A10B-Instruct	122B	untied	248320×3072
	Qwen3.5-397B-A17B-Instruct	397B	untied	248320×4096

Table 8: Llama model inventory and vocabulary-matrix configuration. The Llama-3.x (Llama Team, AI at Meta, 2024) group contrasts small-scale tied with larger-scale untied design switches.

Model Group	Model	Total Params	Status	Vocab Matrix Dim
Llama-3.x	Llama-3.1-70B-Base	70B	untied	128256×8192
	Llama-3.1-70B-Instruct	70B	untied	128256×8192
	Llama-3.1-8B	8B	untied	128256×4096
	Llama-3.1-8B-Instruct	8B	untied	128256×4096

Continued on next page...

Continued: Llama model inventory and vocabulary-matrix configuration.

Model Group	Model	Total Params	Status	Vocab Matrix Dim
	Llama-3.2-1B	1B	tied	128256×2048
	Llama-3.2-1B-Instruct	1B	tied	128256×2048
	Llama-3.2-3B	3B	tied	128256×3072
	Llama-3.2-3B-Instruct	3B	tied	128256×3072

Table 10: Gemma model inventory and vocabulary-matrix configuration. The Gemma family (Gemma Team, 2024, 2025; Google DeepMind, 2026) overall favors tied weights and is an important contrast to Llama/Qwen.

Model Group	Model	Total Params	Status	Vocab Matrix Dim
Gemma-2	Gemma-2-2B	2B	tied	256000×2304
	Gemma-2-2B-Instruct	2B	tied	256000×2304
	Gemma-2-9B	9B	tied	256000×3584
	Gemma-2-9B-Instruct	9B	tied	256000×3584
	Gemma-2-27B	27B	tied	256000×4608
	Gemma-2-27B-Instruct	27B	tied	256000×4608
Gemma-3	Gemma-3-1B	1B	tied	262144×1152
	Gemma-3-1B-Instruct	1B	tied	262144×1152
	Gemma-3-4B	4B	tied	262208×2560
	Gemma-3-4B-Instruct	4B	tied	262208×2560
	Gemma-3-12B	12B	tied	262208×3840
	Gemma-3-12B-Instruct	12B	tied	262208×3840
	Gemma-3-27B	27B	tied	262208×5376
	Gemma-3-27B-Instruct	27B	tied	262208×5376
Gemma-4	Gemma-4-E2B	2B	tied	262144×1536
	Gemma-4-E4B	4B	tied	262144×2560
	Gemma-4-26B-A4B	26B	tied	262144×2816
	Gemma-4-26B-A4B-Instruct	26B	tied	262144×2816
	Gemma-4-31B	31B	tied	262144×5376
	Gemma-4-E2B-Instruct	2B	tied	262144×1536
	Gemma-4-E4B-Instruct	4B	tied	262144×2560
	Gemma-4-31B-Instruct	31B	tied	262144×5376

Gemma exclusions for static spectral analysis The Gemma-4 family and Gemma-3-1B / Gemma-3-1B-Instruct are retained in the inventory to document tied/untied configurations among public models, but they are excluded from the main static SVD/spectral analysis. In the public Gemma-4 weights, input embeddings exhibit a clear row-norm gauge, so raw Euclidean and uncentered spectral statistics are affected by radial scale. Gemma-3-1B shows a similar small-model normalization exception in the Base-Instruct and static-spectrum checks. To keep the spectral comparison on a consistent convention, the main analysis uses only comparable Gemma-2 and Gemma-3 4B/12B/27B models.

Table 12: DeepSeek model inventory and vocabulary-matrix configuration. The DeepSeek group (DeepSeek-AI, 2024a,b, 2025, 2026) includes MoE, reasoning, and distilled models; R1-Distill denotes distilling R1 reasoning capability onto Qwen/Llama target backbones, not DeepSeek-owned small base models. Total parameters for R1-Distill rows are converted from actual safetensors weights.

Model Group	Model	Total Params	Status	Vocab Matrix Dim
DeepSeek-V2 MoE	DeepSeek-V2-Lite	16B	untied	102400×2048
	DeepSeek-V2-Lite-Chat	16B	untied	102400×2048
DeepSeek-V3 MoE	DeepSeek-V3-Base	671B	untied	129280×7168
	DeepSeek-V3	671B	untied	129280×7168
	DeepSeek-V3.1-Base	671B	untied	129280×7168
	DeepSeek-V3.1	671B	untied	129280×7168
	DeepSeek-V3.2	671B	untied	129280×7168
	DeepSeek-V3.2-Speciale	671B	untied	129280×7168

Continued on next page...

Continued: DeepSeek model inventory and vocabulary-matrix configuration.

Model Group	Model	Total Params	Status	Vocab Matrix Dim
DeepSeek-V4 MoE	DeepSeek-V4-Flash	284B	untied	129280×4096
	DeepSeek-V4-Pro	1.6T	untied	129280×7168
DeepSeek-R1	DeepSeek-R1	671B	untied	129280×7168
	DeepSeek-R1-0528	671B	untied	129280×7168
R1-Distill-Qwen (R1→Qwen)	DeepSeek-R1-Distill-Qwen-1.5B	1.8B	untied	151936×1536
	DeepSeek-R1-Distill-Qwen-7B	7.6B	untied	152064×3584
	DeepSeek-R1-Distill-Qwen-14B	15B	untied	152064×5120
	DeepSeek-R1-Distill-Qwen-32B	33B	untied	152064×5120
R1-Distill-Llama (R1→Llama)	DeepSeek-R1-Distill-Llama-8B	8.0B	untied	128256×4096
	DeepSeek-R1-Distill-Llama-70B	71B	untied	128256×8192

DeepSeek-V4 exclusions for static spectral analysis DeepSeek-V4-Flash / Pro are also retained in the inventory, but are excluded from the main untied static-spectrum analysis. Their output path combines HC state mixing, hc_head, RMSNorm, and head.weight; the bare head.weight is therefore not equivalent to the conventional complete LM head used in the other models. Consequently, DeepSeek-V4’s bare U spectrum is not directly comparable to standard untied output-unembedding spectra, and the main analysis treats it as an architectural exception.

Table 14: GLM and MiniMax model inventory and vocabulary-matrix configuration. The GLM (GLM-5 Team, 2026; Z.AI, 2026) and MiniMax (MiniMax, 2026a,b) groups include additional large-scale MoE families to supplement tied/untied choices across vendors.

Model Group	Model	Total Params	Status	Vocab Matrix Dim
MiniMax-M2.5	MiniMax-M2.5-Chat	230B	untied	200064×3072
MiniMax-M2.7	MiniMax-M2.7	229B	untied	200064×3072
GLM-5	GLM-5-744B	744B	untied	154880×6144
GLM-5.1	GLM-5.1	754B	untied	154880×6144

B.2 Hyperparameter Configuration

For global geometric analysis (Protocol I), we use a stratified sampling strategy. Detailed settings are as follows:

- **Matrix objects:** Each model analyzes input embedding matrix E and output unembedding matrix U ; Base-Instruct comparisons compute E_{base} vs. E_{instruct} and U_{base} vs. U_{instruct} separately.
- **Token alignment:** Same-tokenizer/same-vocab models align by token ID; cross-tokenizer or different-vocab models align by token-string intersection; exclude special tokens such as bos/eos/pad/unk; comparisons with fewer than 1000 shared tokens are skipped.
- **Token sample size (N_{tok}):** 20,000 (upper bound). For cross-tokenizer comparisons, this equals the string-intersection size.
- **Pairwise estimate count (N_{pair}):** 5,000,000.
- **GCorr metrics:** For the same batch of token pairs, compute cosine similarity, Euclidean distance, and squared Euclidean distance vectors simultaneously and obtain their Pearson correlations separately.
- **Bootstrap iterations:** $N = 10$ for Base-Instruct comparisons and $N = 100$ for all others.
- **Confidence intervals:** 95% CI based on bootstrap percentiles.

For affine-mapping analysis (Protocol II), fitting uses ordinary least squares (OLS) on a deterministic subset of $N_{\text{fit}} = 24,000$ tokens.

Static spectral subsets. The static vocabulary-matrix spectral analysis applies economy SVD to vocabulary matrices in public model weights and reports mean-axis strength, raw rank-1 energy, centered rank-1 energy, and related participation-ratio statistics. To keep spectra comparable, the main untied spectral subset contains 48 public models and excludes DeepSeek-V4-Flash/Pro because their output head is not a conventional LM-head matrix in the same sense as the other models. The main tied spectral subset contains 34 public models and excludes Gemma-4 variants and

Gemma-3-1B because their row-norm/gauge behavior produces a different normalization regime for the static-spectrum comparison. These exclusions affect only the static spectral panels and do not affect the 37 tied-untied GCorr pairs used as the main evidence that tied matrices are closer to untied U than to untied E .

B.3 Pretraining Experiment Details

This section supplements pretraining experiment settings compressed in Section 5. Experiments cover only pretraining and exclude SFT, RL, DPO, or other post-training pipelines. All variants use the same lightweight dense decoder-only Transformer backbone with hidden size $d = 768$; low-rank variants uniformly use rank $r = 32$ with `lm_head_bias` enabled. Model architecture follows the Qwen3 small dense line (Yang et al., 2025); experiment implementation and the first pretraining dataset come from the MiniMind (Gong, 2025) open-source project, which offers a mature small-model from-scratch training pipeline, low reproduction cost, and high-quality pretraining data.

B.3.1 Variants and Parameter Overhead

To define these variants precisely, let vocabulary size be V , hidden dimension be d , and the shared tied input embedding/output unembedding matrix be $W \in \mathbb{R}^{V \times d}$. For input token x_t and final Transformer hidden state h_t , we denote the effective input matrix by E_{eff} and the effective output matrix by U_{eff} :

$$e_t = E_{\text{eff}}[x_t], \quad z_t = U_{\text{eff}}h_t + b. \quad (20)$$

where $b \in \mathbb{R}^V$ is the output bias enabled uniformly across all variants.

where $\mathbf{1} \in \mathbb{R}^V$; $\beta, \beta_E, \beta_U \in \mathbb{R}^d$; additive LoRA parameters satisfy $A_* \in \mathbb{R}^{V \times r}$ and $B_* \in \mathbb{R}^{r \times d}$; multiplicative LoRA parameters satisfy $P_* \in \mathbb{R}^{d \times r}$ and $Q_* \in \mathbb{R}^{r \times d}$, together with a scalar scale $s > 0$. S3/S11 compare input-side versus output-side bias correction; S4/S5 and S6/S7 compare vocabulary-dimensional additive LoRA versus hidden-dimensional multiplicative LoRA on input/output sides; S8–S10 distinguish whether gains come from alleviating tied constraints or from additional reparameterization capacity of input embeddings themselves.

Let vocabulary size be V , hidden size be $H = 768$, and low-rank rank be $r = 32$. MiniMind and FineWeb-Edu/GPT-2 main pretraining experiments use $s = 1$ for all multiplicative LoRA vari-

Table 16: Overview of S1–S12 embedding-head variants.

Variant	tied?	Description
S1	yes	Standard tied baseline; input embedding shared with <code>lm_head.weight</code>
S2	no	Standard untied baseline; input embedding independent of output head
S3	yes	Input-side hidden-dimensional bias correction, $e - \beta$
S4	yes	Input-side vocabulary-dimensional additive LoRA
S5	yes	Output-side vocabulary-dimensional additive LoRA
S6	yes	Input-side hidden-dimensional multiplicative LoRA
S7	yes	Output-side hidden-dimensional multiplicative LoRA
S8	no	untied + input-side hidden-dimensional bias correction
S9	no	untied + input-side vocabulary-dimensional additive LoRA
S10	no	untied + input-side hidden-dimensional multiplicative LoRA
S11	yes	Output-side hidden-dimensional bias correction
S12	yes	A-LORA (input-side affine LoRA)

Table 17: Effective input and output matrices for S1–S12 model variants.

Model	Relaxation	E_{eff}	U_{eff}
S1	Standard tied (no relaxation)	W	W
S2	Full untied (complete relaxation)	E	U
S3	Input-side bias correction	$W - \mathbf{1}\beta^\top$	W
S4	Input-side standard additive LoRA relaxation	$W + A_{\text{in}}B_{\text{in}}$	W
S5	Output-side standard additive LoRA relaxation	W	$W + A_{\text{out}}B_{\text{out}}$
S6	Input-side multiplicative LoRA	$W(I_d + sP_{\text{in}}Q_{\text{in}})$	W
S7	Output-side multiplicative LoRA	W	$W(I_d + sP_{\text{out}}Q_{\text{out}})$
S8	untied input-side bias correction	$E - \mathbf{1}\beta_E^\top$	U
S9	untied input-side standard additive LoRA relaxation	$E + A_E B_E$	U
S10	untied input-side multiplicative LoRA	$E(I_d + P_E Q_E)$	U
S11	Output-side bias correction	W	W ; logits use $W(h - \beta_U) + b$
S12	A-LORA (input-side affine LoRA)	$W(I_d + sP_{\text{in}}Q_{\text{in}}) - \mathbf{1}\beta^\top$	W

ants (S6/S7/S12); s adds no matrix parameters and only modulates correction strength. In this experiment, hidden-dimensional bias correction adds only H parameters; vocabulary-dimensional additive LoRA adds $Vr + Hr$ parameters; hidden-dimensional multiplicative LoRA adds $2Hr$ parameters; S12-A-LORA adds $H + 2Hr$ parameters. Table 18 gives parameter-overhead ratios relative to the S1 baseline. Vocabulary-dimensional LoRA cost grows linearly with V , so under the GPT-2 vocabulary S4/S5 overhead is much higher than with the MiniMind tokenizer; hidden-dimensional multiplicative LoRA and S12-A-LORA depend only on H and r and do not grow with vocabulary size. In implementation, S12-A-LORA adds both hidden-dimensional bias correction and multiplicative LoRA residual on input lookup vectors (equivalent to the matrix-level affine form $W(I_d + sP_{\text{in}}Q_{\text{in}}) - \mathbf{1}\beta^\top$ with $s = 1$ in the main text); S11 output-side bias correction acts on hidden dimensions, distinct from independent `lm_head.bias`, which is enabled in all variants.

B.3.2 Data and Training Configuration

The FineWeb-Edu/GPT-2 experiment additionally records input embedding/output unembedding gradients for analyzing gradient paths in tied and untied architectures. Recorded quantities include input-embedding-path gradients, output-unembedding-path gradients, total gradients on shared weights in tied variants, and gradients of added modules such as S3/S4/S6/S7/S11/S12. The gradient figures in the main text and appendix are aggregated from the three random seeds.

Validation uses the same tokenizer as training; padding positions in labels are set to -100 to ignore loss. All tables report validation cross-entropy loss over three random seeds and the corresponding PPL converted from the loss.

B.3.3 Full Three-Seed Validation Results

The two tables above report the complete three-seed validation results referenced in the main text and appendix figures.

B.4 Complete Model Comparison List

Due to main-text space limits, we list the supplementary GCorr results. The table includes Base–

Table 18: Added parameter counts relative to the S1 baseline. S1 parameter counts are 63.9M (MiniMind tokenizer) and 97.6M (GPT-2 tokenizer).

Module/Variant	MiniMind Overhead	MiniMind Ratio	GPT-2 Overhead	GPT-2 Ratio
S3/S11 hidden β	768	0.0012%	768	0.0008%
S4/S5 vocab-dimensional additive LoRA	229,376	0.3589%	1,632,800	1.6722%
S6/S7 hidden multiplicative LoRA	49,152	0.0769%	49,152	0.0503%
S12-A-LORA	49,920	0.0781%	49,920	0.0511%
S2 untied head weights	4,915,200	7.6898%	38,597,376	39.5284%
S9 untied + vocab-dimensional LoRA	5,144,576	8.0486%	40,230,176	41.2006%
S10 untied + hidden multiplicative LoRA	4,964,352	7.7667%	38,646,528	39.5788%

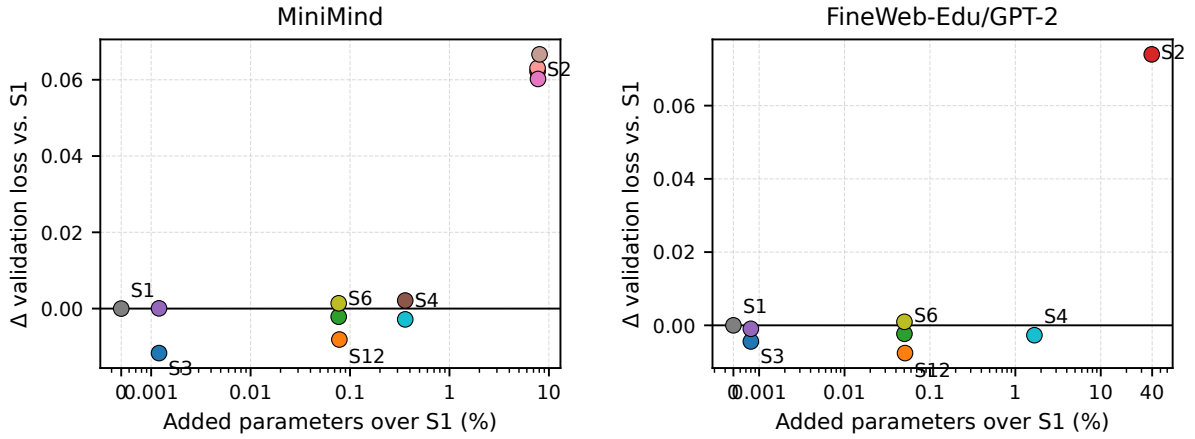


Figure 3: **Relationship between added-parameter ratio and validation loss change.** The x-axis shows added-parameter ratio relative to the S1 tied baseline (log scale; S1 appears near zero); the y-axis shows mean validation loss difference relative to S1 averaged over three seeds. S12-A-LORA lies in the low-overhead region on both datasets and achieves better validation loss than S1; by contrast, S2 full untied introduces large extra parameters but is clearly worse than S1 under the current small-model pretraining setting.

Instruct comparisons and excludes entries involving Gemma-4, Gemma-3-1B, and DeepSeek-V4; the reasons for these exclusions are documented in the model-inventory notes above. The main text uses the focused diagnostic protocol described in Section 3.

Table 19: Pretraining experiment configuration for MiniMind original data and FineWeb-Edu/GPT-2.

Setting	MiniMind Original Data	FineWeb-Edu/GPT-2
Training data	MiniMind original pretraining data (Gong, 2025)	FineWeb-Edu pretraining corpus ($\approx$ 6B tokens) (Penedo et al., 2024)
Tokenizer	MiniMind tokenizer	GPT-2 tokenizer (Radford et al., 2019)
Sequence length	340	340
Train/val split	First 99% / last 1% (val start start_index=8,384,138)	First 99% / last 1% (val start start_index=17,470,587)
Validation sequences	84,689	176,471
Validation tokens	27,069,543	59,766,669
Training variants	S1–S12	S1,S2,S3,S4,S6,S7,S11,S12
Model	Qwen3-aligned dense decoder, 8 layers, $d = 768$	Qwen3-aligned dense decoder, 8 layers, $d = 768$
Low-rank rank	32	32
lm_head_bias	on	on
GPU/DDP	8 GPU, torchrun	8 GPU, torchrun
Batch size	224 per step	80 per step
Effective batch	1,792 sequences	640 sequences
Epochs	2	1
Learning rate	5×10^{-4}	5×10^{-4}
Precision	bfloat16	bfloat16
Seeds	42, 123, 2026	42, 123, 2026

Table 20: Three-seed validation loss for S1–S12 on MiniMind original data. Sorted by mean validation loss from low to high.

Variant	seed42	seed123	seed2026	mean	std	mean PPL	Δ vs S1	Δ vs S2
S3	1.904187	1.908604	1.897083	1.903291	0.005812	6.707934	-0.011671	-0.073830
S12	1.906864	1.903095	1.910587	1.906849	0.003746	6.731843	-0.008113	-0.070273
S4	1.915157	1.904686	1.916560	1.912134	0.006488	6.767515	-0.002828	-0.064987
S6	1.919072	1.906038	1.913362	1.912824	0.006534	6.772186	-0.002138	-0.064297
S1	1.918825	1.913490	1.912571	1.914962	0.003377	6.786681	0.000000	-0.062159
S11	1.915081	1.915418	1.914574	1.915024	0.000425	6.787102	0.000062	-0.062097
S7	1.925734	1.908932	1.914385	1.916350	0.008572	6.796107	0.001388	-0.060771
S5	1.918207	1.927948	1.905078	1.917078	0.011477	6.801057	0.002116	-0.060044
S10	1.970678	1.974093	1.980755	1.975175	0.005125	7.207881	0.060213	-0.001946
S2	1.981365	1.979456	1.970543	1.977121	0.005776	7.221921	0.062159	0.000000
S8	1.981351	1.975633	1.977136	1.978040	0.002964	7.228561	0.063078	0.000919
S9	1.978650	1.981286	1.984940	1.981625	0.003159	7.254522	0.066663	0.004504

Table 21: Three-seed validation loss for key variants under FineWeb-Edu/GPT-2. Sorted by mean validation loss from low to high.

Variant	seed42	seed123	seed2026	mean	std	mean PPL	Δ vs S1	Δ vs S2
S12	3.228119	3.231683	3.231145	3.230316	0.001921	25.287638	-0.007581	-0.081583
S3	3.230755	3.234913	3.234813	3.233494	0.002372	25.368130	-0.004403	-0.078405
S4	3.235644	3.234921	3.235037	3.235201	0.000388	25.411471	-0.002696	-0.076698
S6	3.232097	3.238262	3.236377	3.235579	0.003159	25.421078	-0.002318	-0.076320
S11	3.233531	3.241727	3.235599	3.236952	0.004262	25.456022	-0.000944	-0.074946
S1	3.237538	3.240631	3.235520	3.237896	0.002574	25.480064	0.000000	-0.074002
S7	3.238060	3.240361	3.238147	3.238856	0.001304	25.504528	0.000960	-0.073043
S2	3.312309	3.311959	3.311428	3.311899	0.000444	27.437170	0.074002	0.000000

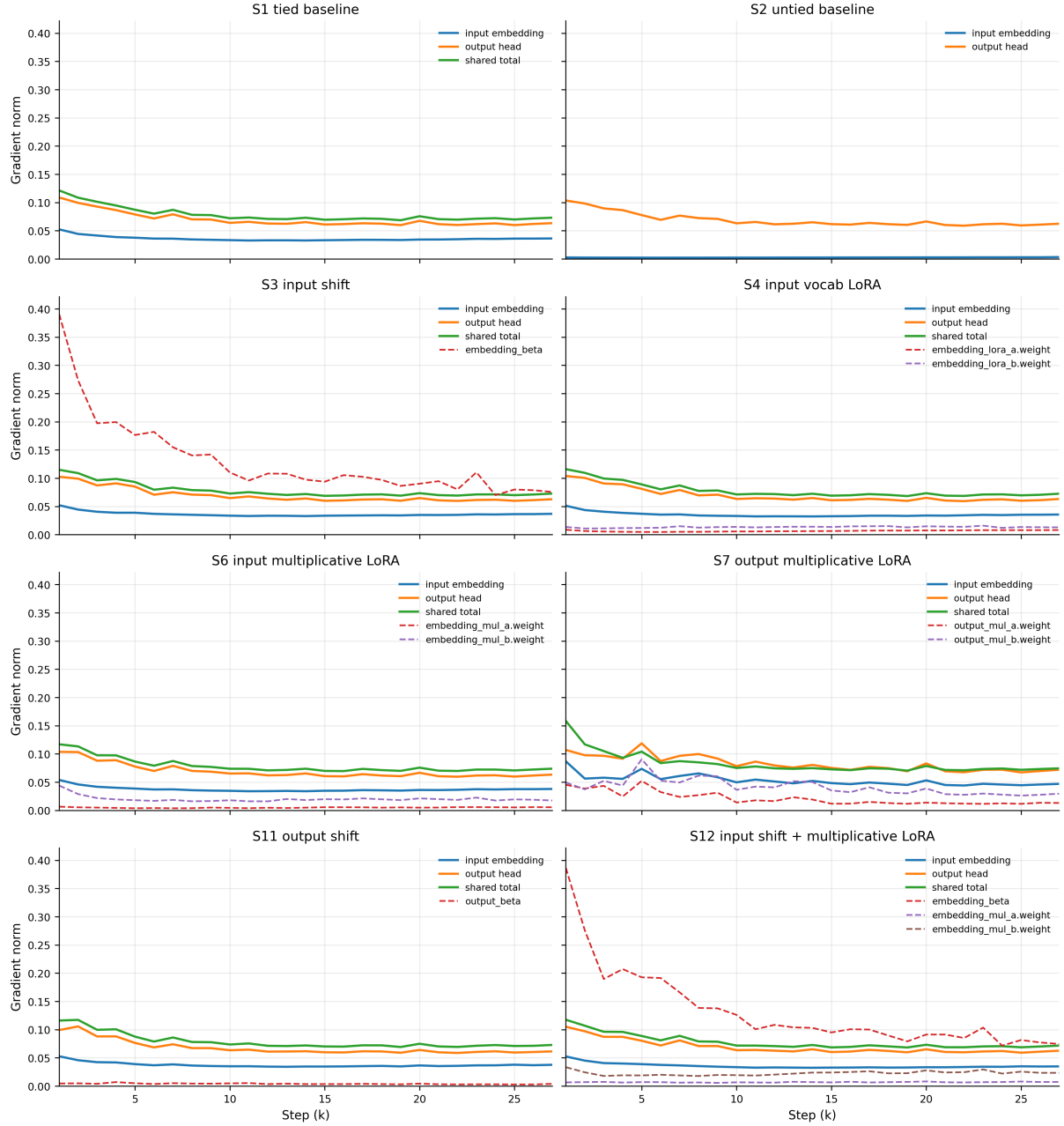


Figure 4: **Gradient-norm overview of baselines and structural variants during FineWeb-Edu/GPT-2 pretraining.** Curves are averaged over seeds 42, 123, and 2026; all subplots share the same y-axis range. Input embedding denotes gradients on input embedding weights from the input path; output head denotes output unembedding-path gradients; shared total denotes total gradients received by the shared vocabulary matrix in tied variants; dashed lines denote gradients of added modules. In the S2 untied baseline, input embedding gradients are much weaker than output unembedding gradients, whereas S3/S12-A-LoRA `embedding_beta` gradients are strong, indicating that the input-side bias module carries a major update signal.

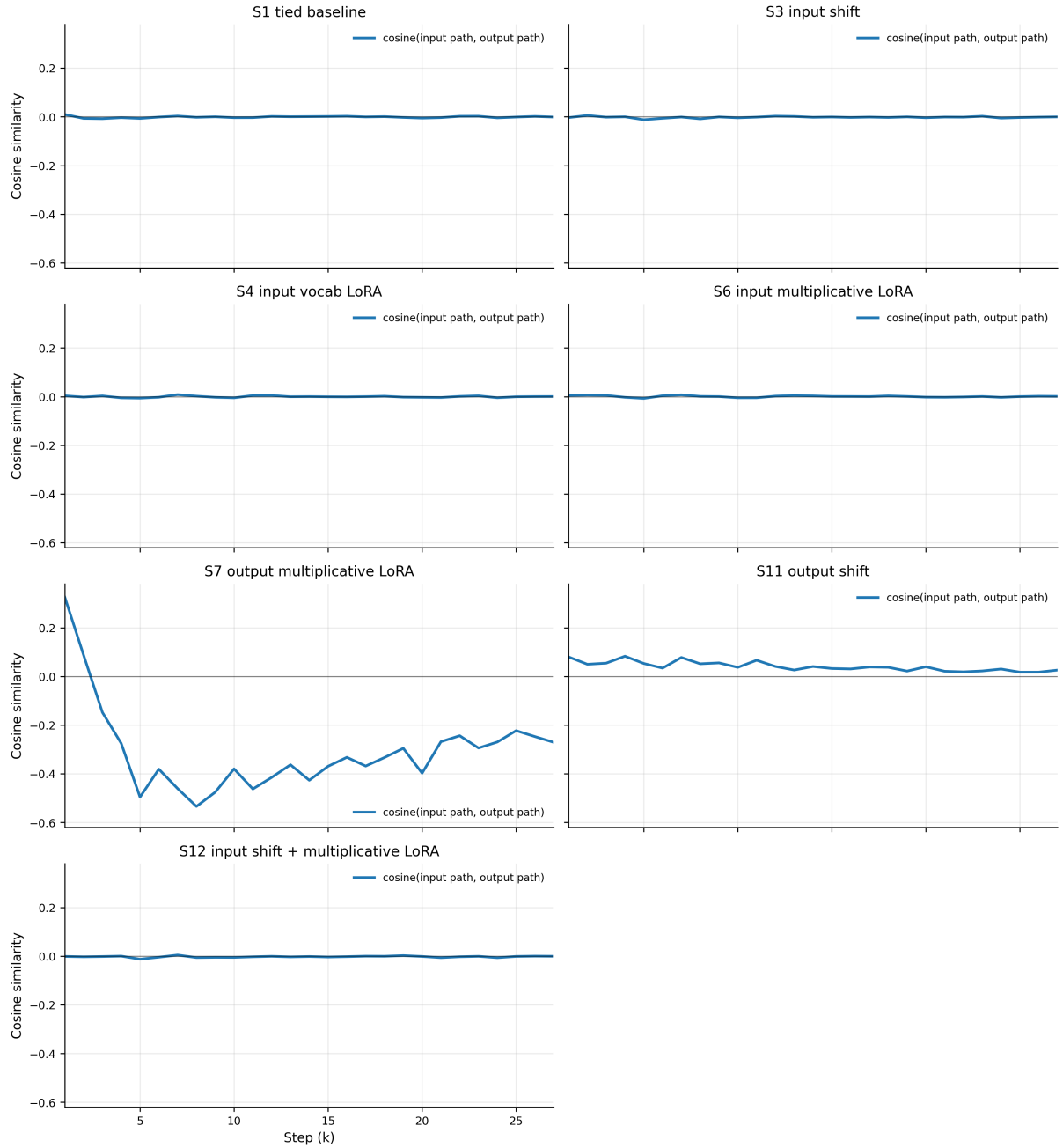


Figure 5: **Input-path versus output-path gradient directions for tied variants during FineWeb-Edu/GPT-2 pretraining.** Curves are averaged over seeds 42, 123, and 2026. Tied split cosine computes cosine similarity between input-embedding-path and output-unembedding-path gradients on the shared vocabulary matrix: near 0 means approximately orthogonal paths, positive means similar directions, negative means conflicting directions. S1/S3/S4/S6/S12 are mostly near 0, indicating no obvious systematic conflict between the two update paths on shared weights; S7 stays negative for long periods at $s = 1$, suggesting output-side multiplicative LoRA makes output-path and input-path gradients clearly oppose each other in shared weight space, which may explain its weaker performance than the S1 baseline.

Table 22: Supplementary GCorr results. We report Pearson correlation (ρ) for Cosine and Euclidean GCorr; squared Euclidean GCorr is computed with the same protocol but omitted from this compact table.

Task	Model Pair (A vs. B)	$E_{\cos}$	E_{euc}	$U_{\cos}$	U_{euc}
Base/Instruct	Qwen3-0.6B-Base vs. Qwen3-0.6B	0.990	0.994	0.990	0.994
	Qwen3-1.7B-Base vs. Qwen3-1.7B	0.995	0.998	0.995	0.998
	Qwen3-4B-Base vs. Qwen3-4B	0.997	0.997	0.997	0.997
	Qwen3-8B-Base vs. Qwen3-8B	0.994	1.000	0.998	0.999
	Qwen3-14B-Base vs. Qwen3-14B	0.993	1.000	0.997	0.999
	Qwen3-30B-A3B-Base vs. Qwen3-30B-A3B	0.987	1.000	0.998	0.998
	Qwen3.5-0.8B-Base vs. Qwen3.5-0.8B-Instruct	0.999	0.999	0.999	0.999
	Qwen3.5-2B-Base vs. Qwen3.5-2B-Instruct	1.000	0.999	1.000	0.999
	Qwen3.5-4B-Base vs. Qwen3.5-4B-Instruct	1.000	0.999	1.000	0.999
	Qwen3.5-9B-Base vs. Qwen3.5-9B-Instruct	0.998	1.000	1.000	0.999
	Qwen3.5-35B-A3B-Base vs. Qwen3.5-35B-A3B-Instruct	0.999	1.000	1.000	0.999
	Qwen2.5-0.5B vs. Qwen2.5-0.5B-Instruct	0.998	0.996	0.998	0.996
	Qwen2.5-1.5B vs. Qwen2.5-1.5B-Instruct	1.000	1.000	1.000	1.000
	Qwen2.5-3B vs. Qwen2.5-3B-Instruct	1.000	1.000	1.000	1.000
	Qwen2.5-7B vs. Qwen2.5-7B-Instruct	1.000	1.000	1.000	1.000
	Qwen2.5-14B vs. Qwen2.5-14B-Instruct	1.000	1.000	0.999	1.000
	Qwen2.5-32B vs. Qwen2.5-32B-Instruct	1.000	1.000	1.000	1.000
	Qwen2.5-72B-Base vs. Qwen2.5-72B-Instruct	1.000	1.000	1.000	1.000
	Llama-3.2-1B vs. Llama-3.2-1B-Instruct	0.981	0.970	0.981	0.970
	Llama-3.2-3B vs. Llama-3.2-3B-Instruct	0.986	0.985	0.986	0.985
	Llama-3.1-8B vs. Llama-3.1-8B-Instruct	0.998	1.000	0.999	0.999
	Llama-3.1-70B-Base vs. Llama-3.1-70B-Instruct	0.999	1.000	0.998	0.999
	Gemma-2-2B vs. Gemma-2-2B-Instruct	1.000	0.999	1.000	0.999
	Gemma-2-9B vs. Gemma-2-9B-Instruct	0.999	0.996	0.999	0.996
	Gemma-2-27B vs. Gemma-2-27B-Instruct	0.998	0.982	0.998	0.982
	Gemma-3-4B vs. Gemma-3-4B-Instruct	0.979	0.894	0.979	0.894
	Gemma-3-12B vs. Gemma-3-12B-Instruct	0.988	0.946	0.988	0.946
	Gemma-3-27B vs. Gemma-3-27B-Instruct	0.990	0.953	0.990	0.953
	DeepSeek-V3-Base vs. DeepSeek-V3	1.000	1.000	1.000	1.000
	DeepSeek-V3.1-Base vs. DeepSeek-V3.1	1.000	1.000	1.000	1.000
Model Series	Llama-3.1-70B-Instruct vs. Llama-3.1-8B-Instruct	0.410	0.885	0.843	0.897
	Llama-3.1-70B-Instruct vs. Llama-3.2-1B-Instruct	0.178	0.354	0.752	0.645
	Llama-3.1-70B-Instruct vs. Llama-3.2-3B-Instruct	0.159	0.434	0.750	0.726
	Llama-3.1-8B-Instruct vs. Llama-3.2-1B-Instruct	0.074	0.359	0.875	0.672
	Llama-3.1-8B-Instruct vs. Llama-3.2-3B-Instruct	0.057	0.424	0.877	0.722
	Llama-3.2-1B-Instruct vs. Llama-3.2-3B-Instruct	0.949	0.959	0.949	0.959
	Qwen2.5-0.5B-Instruct vs. Qwen2.5-1.5B-Instruct	0.851	0.889	0.851	0.889
	Qwen2.5-0.5B-Instruct vs. Qwen2.5-3B-Instruct	0.848	0.862	0.848	0.862
	Qwen2.5-0.5B-Instruct vs. Qwen2.5-7B-Instruct	0.059	0.624	0.849	0.803
	Qwen2.5-0.5B-Instruct vs. Qwen2.5-14B-Instruct	0.150	0.587	0.666	0.753
	Qwen2.5-0.5B-Instruct vs. Qwen2.5-32B-Instruct	0.152	0.560	0.742	0.767
	Qwen2.5-0.5B-Instruct vs. Qwen2.5-72B-Instruct	0.118	0.516	0.825	0.753
	Qwen2.5-1.5B-Instruct vs. Qwen2.5-3B-Instruct	0.915	0.972	0.915	0.972
	Qwen2.5-1.5B-Instruct vs. Qwen2.5-7B-Instruct	0.090	0.620	0.793	0.796
	Qwen2.5-1.5B-Instruct vs. Qwen2.5-14B-Instruct	0.158	0.604	0.736	0.876
	Qwen2.5-1.5B-Instruct vs. Qwen2.5-32B-Instruct	0.166	0.575	0.769	0.852
	Qwen2.5-1.5B-Instruct vs. Qwen2.5-72B-Instruct	0.143	0.514	0.781	0.760
	Qwen2.5-3B-Instruct vs. Qwen2.5-7B-Instruct	0.092	0.665	0.822	0.787
	Qwen2.5-3B-Instruct vs. Qwen2.5-14B-Instruct	0.165	0.653	0.786	0.915
	Qwen2.5-3B-Instruct vs. Qwen2.5-32B-Instruct	0.171	0.623	0.814	0.892
	Qwen2.5-3B-Instruct vs. Qwen2.5-72B-Instruct	0.144	0.556	0.808	0.781
	Qwen2.5-7B-Instruct vs. Qwen2.5-14B-Instruct	0.306	0.971	0.755	0.807
	Qwen2.5-7B-Instruct vs. Qwen2.5-32B-Instruct	0.313	0.962	0.844	0.822
	Qwen2.5-7B-Instruct vs. Qwen2.5-72B-Instruct	0.278	0.938	0.928	0.918
	Qwen2.5-14B-Instruct vs. Qwen2.5-32B-Instruct	0.502	0.993	0.893	0.979
	Qwen2.5-14B-Instruct vs. Qwen2.5-72B-Instruct	0.277	0.950	0.763	0.815
	Qwen2.5-32B-Instruct vs. Qwen2.5-72B-Instruct	0.301	0.955	0.870	0.845

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Task	Model Pair (A vs. B)	$E_{\cos}$	E_{euc}	$U_{\cos}$	U_{euc}
	Qwen3-0.6B vs. Qwen3-1.7B	0.857	0.932	0.857	0.932
	Qwen3-0.6B vs. Qwen3-4B	0.884	0.900	0.884	0.900
	Qwen3-0.6B vs. Qwen3-8B	0.147	0.572	0.846	0.818
	Qwen3-0.6B vs. Qwen3-14B	0.098	0.527	0.733	0.764
	Qwen3-0.6B vs. Qwen3-32B	0.121	0.514	0.741	0.718
	Qwen3-0.6B vs. Qwen3-30B-A3B	0.159	0.594	0.828	0.823
	Qwen3-0.6B vs. Qwen3-235B-A22B	0.149	0.522	0.683	0.745
	Qwen3-1.7B vs. Qwen3-4B	0.874	0.967	0.874	0.967
	Qwen3-1.7B vs. Qwen3-8B	0.140	0.659	0.884	0.943
	Qwen3-1.7B vs. Qwen3-14B	0.099	0.612	0.775	0.896
	Qwen3-1.7B vs. Qwen3-32B	0.120	0.597	0.760	0.845
	Qwen3-1.7B vs. Qwen3-30B-A3B	0.158	0.720	0.849	0.915
	Qwen3-1.7B vs. Qwen3-235B-A22B	0.151	0.626	0.766	0.877
	Qwen3-4B vs. Qwen3-8B	0.130	0.697	0.919	0.966
	Qwen3-4B vs. Qwen3-14B	0.073	0.652	0.811	0.923
	Qwen3-4B vs. Qwen3-32B	0.096	0.631	0.819	0.878
	Qwen3-4B vs. Qwen3-30B-A3B	0.125	0.744	0.875	0.940
	Qwen3-4B vs. Qwen3-235B-A22B	0.121	0.658	0.745	0.914
	Qwen3-8B vs. Qwen3-14B	0.347	0.988	0.832	0.954
	Qwen3-8B vs. Qwen3-32B	0.353	0.979	0.819	0.903
	Qwen3-8B vs. Qwen3-30B-A3B	0.291	0.800	0.884	0.950
	Qwen3-8B vs. Qwen3-235B-A22B	0.267	0.911	0.793	0.955
	Qwen3-14B vs. Qwen3-32B	0.538	0.994	0.919	0.982
	Qwen3-14B vs. Qwen3-30B-A3B	0.309	0.747	0.768	0.888
	Qwen3-14B vs. Qwen3-235B-A22B	0.290	0.890	0.823	0.933
	Qwen3-32B vs. Qwen3-30B-A3B	0.321	0.725	0.794	0.838
	Qwen3-32B vs. Qwen3-235B-A22B	0.305	0.882	0.791	0.896
	Qwen3-30B-A3B vs. Qwen3-235B-A22B	0.381	0.879	0.751	0.946
	Qwen3.5-0.8B-Instruct vs. Qwen3.5-2B-Instruct	0.932	0.916	0.932	0.916
	Qwen3.5-0.8B-Instruct vs. Qwen3.5-4B-Instruct	0.915	0.874	0.915	0.874
	Qwen3.5-0.8B-Instruct vs. Qwen3.5-9B-Instruct	0.180	0.427	0.874	0.722
	Qwen3.5-0.8B-Instruct vs. Qwen3.5-27B-Instruct	0.243	0.468	0.859	0.672
	Qwen3.5-2B-Instruct vs. Qwen3.5-4B-Instruct	0.949	0.959	0.949	0.959
	Qwen3.5-2B-Instruct vs. Qwen3.5-9B-Instruct	0.157	0.377	0.895	0.828
	Qwen3.5-2B-Instruct vs. Qwen3.5-27B-Instruct	0.227	0.414	0.880	0.789
	Qwen3.5-4B-Instruct vs. Qwen3.5-9B-Instruct	0.162	0.423	0.923	0.890
	Qwen3.5-4B-Instruct vs. Qwen3.5-27B-Instruct	0.231	0.444	0.912	0.862
	Qwen3.5-9B-Instruct vs. Qwen3.5-27B-Instruct	0.408	0.962	0.946	0.985
	Qwen3.5-35B-A3B-Instruct vs.	0.412	0.936	0.904	0.963
	Qwen3.5-122B-A10B-Instruct				
	Qwen3.5-35B-A3B-Instruct vs.	0.414	0.931	0.831	0.914
	Qwen3.5-397B-A17B-Instruct				
	Qwen3.5-122B-A10B-Instruct vs.	0.499	0.967	0.818	0.942
	Qwen3.5-397B-A17B-Instruct				
	Gemma-2-2B-Instruct vs. Gemma-2-9B-Instruct	0.697	0.762	0.697	0.762
	Gemma-2-2B-Instruct vs. Gemma-2-27B-Instruct	0.392	0.105	0.392	0.104
	Gemma-2-9B-Instruct vs. Gemma-2-27B-Instruct	0.368	0.091	0.368	0.091
	Gemma-3-4B-Instruct vs. Gemma-3-12B-Instruct	0.756	0.864	0.756	0.864
	Gemma-3-4B-Instruct vs. Gemma-3-27B-Instruct	0.711	0.831	0.712	0.831
	Gemma-3-12B-Instruct vs. Gemma-3-27B-Instruct	0.834	0.916	0.835	0.916
	MiniMax-M2.5-Chat vs. MiniMax-M2.7	0.992	0.997	0.976	0.985
	GLM-5-744B vs. GLM-5.1	0.996	1.000	1.000	1.000
	DeepSeek-V2-Lite vs. DeepSeek-V2-Lite-Chat	1.000	1.000	1.000	1.000
	DeepSeek-V3-Base vs. DeepSeek-V3	1.000	1.000	1.000	1.000
	DeepSeek-V3-Base vs. DeepSeek-V3.1-Base	1.000	1.000	1.000	1.000
	DeepSeek-V3-Base vs. DeepSeek-V3.1	1.000	1.000	1.000	1.000
	DeepSeek-V3-Base vs. DeepSeek-V3.2	1.000	1.000	1.000	1.000
	DeepSeek-V3-Base vs. DeepSeek-V3.2-Speciale	1.000	1.000	1.000	1.000
	DeepSeek-V3 vs. DeepSeek-V3.1-Base	1.000	1.000	1.000	1.000
	DeepSeek-V3 vs. DeepSeek-V3.1	1.000	1.000	1.000	1.000
	DeepSeek-V3 vs. DeepSeek-V3.2	1.000	1.000	1.000	1.000
	DeepSeek-V3 vs. DeepSeek-V3.2-Speciale	1.000	1.000	1.000	1.000
	DeepSeek-V3.1-Base vs. DeepSeek-V3.1	1.000	1.000	1.000	1.000

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Task	Model Pair (A vs. B)	$E_{\cos}$	E_{euc}	$U_{\cos}$	U_{euc}
	DeepSeek-V3.1-Base vs. DeepSeek-V3.2	1.000	1.000	1.000	1.000
	DeepSeek-V3.1-Base vs. DeepSeek-V3.2-Speciale	1.000	1.000	1.000	1.000
	DeepSeek-V3.1 vs. DeepSeek-V3.2	1.000	1.000	1.000	1.000
	DeepSeek-V3.1 vs. DeepSeek-V3.2-Speciale	1.000	1.000	1.000	1.000
	DeepSeek-V3.2 vs. DeepSeek-V3.2-Speciale	1.000	1.000	1.000	1.000
	DeepSeek-R1 vs. DeepSeek-R1-0528	1.000	1.000	1.000	1.000
	DeepSeek-R1-Distill-Qwen-1.5B vs. DeepSeek-R1-Distill-Qwen-7B	0.223	0.897	0.953	0.930
	DeepSeek-R1-Distill-Qwen-1.5B vs. DeepSeek-R1-Distill-Qwen-14B	0.154	0.818	0.624	0.519
	DeepSeek-R1-Distill-Qwen-1.5B vs. DeepSeek-R1-Distill-Qwen-32B	0.161	0.809	0.661	0.538
	DeepSeek-R1-Distill-Qwen-7B vs. DeepSeek-R1-Distill-Qwen-14B	0.253	0.867	0.674	0.638
	DeepSeek-R1-Distill-Qwen-7B vs. DeepSeek-R1-Distill-Qwen-32B	0.264	0.872	0.718	0.653
	DeepSeek-R1-Distill-Qwen-14B vs. DeepSeek-R1-Distill-Qwen-32B	0.501	0.993	0.903	0.979
	DeepSeek-R1-Distill-Llama-8B vs. DeepSeek-R1-Distill-Llama-70B	0.405	0.878	0.843	0.888
Cross-Scale	Qwen3-0.6B vs. Qwen2.5-0.5B-Instruct	0.783	0.876	0.783	0.876
	Qwen3-0.6B vs. Qwen2.5-1.5B-Instruct	0.861	0.915	0.861	0.915
	Qwen3-0.6B vs. Qwen2.5-3B-Instruct	0.850	0.892	0.850	0.892
	Qwen3-0.6B vs. Llama-3.2-1B-Instruct	0.623	0.528	0.623	0.528
	Qwen3-0.6B vs. Llama-3.2-3B-Instruct	0.619	0.468	0.619	0.468
	Qwen3-0.6B vs. Gemma-2-2B-Instruct	0.595	0.291	0.595	0.291
	Qwen3-0.6B vs. Gemma-3-4B-Instruct	0.312	0.177	0.312	0.177
	Qwen3-0.6B vs. Qwen3.5-0.8B-Instruct	0.634	0.604	0.634	0.604
	Qwen3-0.6B vs. Qwen3.5-2B-Instruct	0.628	0.599	0.628	0.599
	Qwen3-0.6B vs. Qwen3.5-4B-Instruct	0.589	0.583	0.589	0.583
	Qwen3-1.7B vs. Qwen2.5-0.5B-Instruct	0.723	0.850	0.723	0.850
	Qwen3-1.7B vs. Qwen2.5-1.5B-Instruct	0.848	0.933	0.848	0.933
	Qwen3-1.7B vs. Qwen2.5-3B-Instruct	0.848	0.948	0.848	0.948
	Qwen3-1.7B vs. Llama-3.2-1B-Instruct	0.641	0.617	0.641	0.617
	Qwen3-1.7B vs. Llama-3.2-3B-Instruct	0.628	0.600	0.628	0.600
	Qwen3-1.7B vs. Gemma-2-2B-Instruct	0.619	0.124	0.619	0.124
	Qwen3-1.7B vs. Gemma-3-4B-Instruct	0.364	0.025	0.364	0.025
	Qwen3-1.7B vs. Qwen3.5-0.8B-Instruct	0.623	0.568	0.623	0.568
	Qwen3-1.7B vs. Qwen3.5-2B-Instruct	0.625	0.611	0.625	0.611
	Qwen3-1.7B vs. Qwen3.5-4B-Instruct	0.599	0.628	0.599	0.628
	Qwen3-4B vs. Qwen2.5-0.5B-Instruct	0.820	0.848	0.820	0.848
	Qwen3-4B vs. Qwen2.5-1.5B-Instruct	0.895	0.916	0.895	0.916
	Qwen3-4B vs. Qwen2.5-3B-Instruct	0.898	0.940	0.898	0.940
	Qwen3-4B vs. Llama-3.2-1B-Instruct	0.709	0.714	0.709	0.714
	Qwen3-4B vs. Llama-3.2-3B-Instruct	0.710	0.724	0.710	0.724
	Qwen3-4B vs. Gemma-2-2B-Instruct	0.653	0.133	0.653	0.133
	Qwen3-4B vs. Gemma-3-4B-Instruct	0.283	0.040	0.283	0.040
	Qwen3-4B vs. Qwen3.5-0.8B-Instruct	0.651	0.607	0.651	0.607
	Qwen3-4B vs. Qwen3.5-2B-Instruct	0.665	0.649	0.665	0.649
	Qwen3-4B vs. Qwen3.5-4B-Instruct	0.637	0.677	0.637	0.677
	Qwen2.5-0.5B-Instruct vs. Llama-3.2-1B-Instruct	0.736	0.456	0.736	0.456
	Qwen2.5-0.5B-Instruct vs. Llama-3.2-3B-Instruct	0.726	0.400	0.726	0.400
	Qwen2.5-0.5B-Instruct vs. Gemma-2-2B-Instruct	0.680	0.333	0.680	0.333
	Qwen2.5-0.5B-Instruct vs. Gemma-3-4B-Instruct	0.160	0.190	0.160	0.190
	Qwen2.5-0.5B-Instruct vs. Qwen3.5-0.8B-Instruct	0.722	0.612	0.722	0.612
	Qwen2.5-0.5B-Instruct vs. Qwen3.5-2B-Instruct	0.736	0.644	0.736	0.644
	Qwen2.5-0.5B-Instruct vs. Qwen3.5-4B-Instruct	0.721	0.628	0.721	0.628
	Qwen2.5-1.5B-Instruct vs. Llama-3.2-1B-Instruct	0.752	0.608	0.752	0.608
	Qwen2.5-1.5B-Instruct vs. Llama-3.2-3B-Instruct	0.744	0.567	0.744	0.567
	Qwen2.5-1.5B-Instruct vs. Gemma-2-2B-Instruct	0.704	0.227	0.704	0.227
	Qwen2.5-1.5B-Instruct vs. Gemma-3-4B-Instruct	0.229	0.095	0.229	0.095
	Qwen2.5-1.5B-Instruct vs. Qwen3.5-0.8B-Instruct	0.717	0.585	0.717	0.585
	Qwen2.5-1.5B-Instruct vs. Qwen3.5-2B-Instruct	0.728	0.629	0.728	0.629

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Task	Model Pair (A vs. B)	$E_{\cos}$	E_{euc}	$U_{\cos}$	U_{euc}
	Qwen2.5-1.5B-Instruct vs. Qwen3.5-4B-Instruct	0.707	0.629	0.707	0.629
	Qwen2.5-3B-Instruct vs. Llama-3.2-1B-Instruct	0.760	0.654	0.760	0.654
	Qwen2.5-3B-Instruct vs. Llama-3.2-3B-Instruct	0.751	0.637	0.751	0.637
	Qwen2.5-3B-Instruct vs. Gemma-2-2B-Instruct	0.696	0.151	0.696	0.151
	Qwen2.5-3B-Instruct vs. Gemma-3-4B-Instruct	0.260	0.038	0.260	0.038
	Qwen2.5-3B-Instruct vs. Qwen3.5-0.8B-Instruct	0.699	0.560	0.699	0.560
	Qwen2.5-3B-Instruct vs. Qwen3.5-2B-Instruct	0.711	0.617	0.711	0.617
	Qwen2.5-3B-Instruct vs. Qwen3.5-4B-Instruct	0.702	0.631	0.702	0.631
	Llama-3.2-1B-Instruct vs. Gemma-2-2B-Instruct	0.675	-0.004	0.675	-0.004
	Llama-3.2-1B-Instruct vs. Gemma-3-4B-Instruct	0.181	0.082	0.181	0.082
	Llama-3.2-1B-Instruct vs. Qwen3.5-0.8B-Instruct	0.796	0.716	0.796	0.716
	Llama-3.2-1B-Instruct vs. Qwen3.5-2B-Instruct	0.804	0.733	0.804	0.733
	Llama-3.2-1B-Instruct vs. Qwen3.5-4B-Instruct	0.823	0.745	0.823	0.745
	Llama-3.2-3B-Instruct vs. Gemma-2-2B-Instruct	0.654	-0.028	0.654	-0.027
	Llama-3.2-3B-Instruct vs. Gemma-3-4B-Instruct	0.140	0.029	0.140	0.029
	Llama-3.2-3B-Instruct vs. Qwen3.5-0.8B-Instruct	0.794	0.657	0.794	0.657
	Llama-3.2-3B-Instruct vs. Qwen3.5-2B-Instruct	0.809	0.701	0.809	0.701
	Llama-3.2-3B-Instruct vs. Qwen3.5-4B-Instruct	0.826	0.736	0.826	0.736
	Gemma-2-2B-Instruct vs. Gemma-3-4B-Instruct	0.312	0.220	0.312	0.220
	Gemma-2-2B-Instruct vs. Qwen3.5-0.8B-Instruct	0.704	0.125	0.704	0.125
	Gemma-2-2B-Instruct vs. Qwen3.5-2B-Instruct	0.725	0.108	0.725	0.108
	Gemma-2-2B-Instruct vs. Qwen3.5-4B-Instruct	0.731	0.090	0.731	0.090
	Gemma-3-4B-Instruct vs. Qwen3.5-0.8B-Instruct	0.223	0.187	0.223	0.187
	Gemma-3-4B-Instruct vs. Qwen3.5-2B-Instruct	0.157	0.101	0.157	0.101
	Gemma-3-4B-Instruct vs. Qwen3.5-4B-Instruct	0.168	0.065	0.168	0.065
	Qwen3-8B vs. Qwen2.5-7B-Instruct	0.286	0.941	0.749	0.696
	Qwen3-8B vs. Qwen2.5-14B-Instruct	0.299	0.961	0.780	0.925
	Qwen3-8B vs. Llama-3.1-8B-Instruct	0.285	0.117	0.613	0.713
	Qwen3-8B vs. Gemma-2-9B-Instruct	0.285	-0.280	0.496	-0.115
	Qwen3-8B vs. Gemma-3-12B-Instruct	0.414	-0.078	0.104	-0.151
	Qwen3-8B vs. Qwen3.5-9B-Instruct	0.268	0.733	0.630	0.718
	Qwen3-14B vs. Qwen2.5-7B-Instruct	0.311	0.922	0.763	0.718
	Qwen3-14B vs. Qwen2.5-14B-Instruct	0.322	0.958	0.870	0.952
	Qwen3-14B vs. Llama-3.1-8B-Instruct	0.313	0.059	0.620	0.782
	Qwen3-14B vs. Gemma-2-9B-Instruct	0.278	-0.266	0.450	-0.077
	Qwen3-14B vs. Gemma-3-12B-Instruct	0.449	-0.061	0.227	-0.168
	Qwen3-14B vs. Qwen3.5-9B-Instruct	0.290	0.699	0.541	0.727
	Qwen2.5-7B-Instruct vs. Llama-3.1-8B-Instruct	0.307	0.249	0.716	0.707
	Qwen2.5-7B-Instruct vs. Gemma-2-9B-Instruct	0.286	-0.272	0.520	0.285
	Qwen2.5-7B-Instruct vs. Gemma-3-12B-Instruct	0.439	0.010	-0.016	0.047
	Qwen2.5-7B-Instruct vs. Qwen3.5-9B-Instruct	0.285	0.688	0.632	0.583
	Qwen2.5-14B-Instruct vs. Llama-3.1-8B-Instruct	0.274	0.191	0.612	0.816
	Qwen2.5-14B-Instruct vs. Gemma-2-9B-Instruct	0.241	-0.276	0.421	-0.006
	Qwen2.5-14B-Instruct vs. Gemma-3-12B-Instruct	0.372	-0.074	0.222	-0.140
	Qwen2.5-14B-Instruct vs. Qwen3.5-9B-Instruct	0.260	0.688	0.496	0.741
	Llama-3.1-8B-Instruct vs. Gemma-2-9B-Instruct	0.311	0.056	0.459	0.073
	Llama-3.1-8B-Instruct vs. Gemma-3-12B-Instruct	0.548	0.176	-0.007	-0.077
	Llama-3.1-8B-Instruct vs. Qwen3.5-9B-Instruct	0.322	0.527	0.784	0.806
	Gemma-2-9B-Instruct vs. Gemma-3-12B-Instruct	0.261	0.244	0.261	0.244
	Gemma-2-9B-Instruct vs. Qwen3.5-9B-Instruct	0.288	-0.323	0.435	-0.096
	Gemma-3-12B-Instruct vs. Qwen3.5-9B-Instruct	0.474	-0.085	0.002	-0.122
	Qwen3-32B vs. Qwen2.5-32B-Instruct	0.337	0.964	0.894	0.943
	Qwen3-32B vs. Qwen2.5-72B-Instruct	0.289	0.921	0.838	0.723
	Qwen3-32B vs. Llama-3.1-70B-Instruct	0.371	0.059	0.717	0.777
	Qwen3-32B vs. Gemma-2-27B-Instruct	0.282	-0.092	0.220	0.068
	Qwen3-32B vs. Gemma-3-27B-Instruct	0.423	-0.086	0.094	-0.118
	Qwen3-32B vs. Qwen3.5-27B-Instruct	0.300	0.853	0.620	0.744
	Qwen3-235B-A22B vs. Qwen2.5-32B-Instruct	0.285	0.877	0.759	0.909
	Qwen3-235B-A22B vs. Qwen2.5-72B-Instruct	0.311	0.901	0.645	0.732
	Qwen3-235B-A22B vs. Llama-3.1-70B-Instruct	0.423	0.396	0.519	0.716
	Qwen3-235B-A22B vs. Gemma-2-27B-Instruct	0.231	-0.081	0.324	0.036
	Qwen3-235B-A22B vs. Gemma-3-27B-Instruct	0.393	-0.078	0.368	-0.138
	Qwen3-235B-A22B vs. Qwen3.5-27B-Instruct	0.282	0.816	0.496	0.676

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Task	Model Pair (A vs. B)	$E_{\cos}$	E_{euc}	$U_{\cos}$	U_{euc}
	Qwen2.5-32B-Instruct vs. Llama-3.1-70B-Instruct	0.341	0.159	0.671	0.843
	Qwen2.5-32B-Instruct vs. Gemma-2-27B-Instruct	0.241	-0.100	0.214	0.061
	Qwen2.5-32B-Instruct vs. Gemma-3-27B-Instruct	0.350	-0.105	0.083	-0.119
	Qwen2.5-32B-Instruct vs. Qwen3.5-27B-Instruct	0.271	0.831	0.543	0.769
	Qwen2.5-72B-Instruct vs. Llama-3.1-70B-Instruct	0.393	0.212	0.616	0.699
	Qwen2.5-72B-Instruct vs. Gemma-2-27B-Instruct	0.224	-0.107	0.142	0.178
	Qwen2.5-72B-Instruct vs. Gemma-3-27B-Instruct	0.385	-0.067	-0.059	0.041
	Qwen2.5-72B-Instruct vs. Qwen3.5-27B-Instruct	0.258	0.819	0.569	0.579
	Llama-3.1-70B-Instruct vs. Gemma-2-27B-Instruct	0.339	0.109	0.241	0.095
	Llama-3.1-70B-Instruct vs. Gemma-3-27B-Instruct	0.543	0.151	0.045	-0.113
	Llama-3.1-70B-Instruct vs. Qwen3.5-27B-Instruct	0.364	0.325	0.700	0.757
	Gemma-2-27B-Instruct vs. Gemma-3-27B-Instruct	0.209	0.243	0.209	0.243
	Gemma-2-27B-Instruct vs. Qwen3.5-27B-Instruct	0.298	-0.058	0.176	0.086
	Gemma-3-27B-Instruct vs. Qwen3.5-27B-Instruct	0.425	-0.116	-0.033	-0.133
MoE	Qwen3-30B-A3B vs. Qwen3.5-35B-A3B-Instruct	0.379	0.707	0.733	0.766
Cross-Family	Qwen3-30B-A3B vs. Qwen3.6-35B-A3B	0.379	0.710	0.746	0.758
	Qwen3-30B-A3B vs. DeepSeek-V2-Lite-Chat	0.464	0.649	0.572	0.728
	Qwen3-30B-A3B vs. DeepSeek-R1-0528	0.449	0.404	0.382	0.793
	Qwen3-30B-A3B vs. MiniMax-M2.7	0.413	0.425	0.661	0.702
	Qwen3.5-35B-A3B-Instruct vs. Qwen3.6-35B-A3B	0.953	0.997	0.994	0.990
	Qwen3.5-35B-A3B-Instruct vs. DeepSeek-V2-Lite-Chat	0.458	0.527	0.546	0.641
	Qwen3.5-35B-A3B-Instruct vs. DeepSeek-R1-0528	0.462	0.321	0.471	0.695
	Qwen3.5-35B-A3B-Instruct vs. MiniMax-M2.7	0.410	0.199	0.763	0.745
	Qwen3.6-35B-A3B vs. DeepSeek-V2-Lite-Chat	0.458	0.531	0.544	0.614
	Qwen3.6-35B-A3B vs. DeepSeek-R1-0528	0.462	0.320	0.464	0.690
	Qwen3.6-35B-A3B vs. MiniMax-M2.7	0.410	0.204	0.769	0.716
	DeepSeek-V2-Lite-Chat vs. DeepSeek-R1-0528	0.568	0.119	0.449	0.623
	DeepSeek-V2-Lite-Chat vs. MiniMax-M2.7	0.505	0.564	0.481	0.793
	DeepSeek-R1-0528 vs. MiniMax-M2.7	0.549	-0.184	0.484	0.560

B.4.1 Task1 Base-Instruct input-side GCorr summary (Insight 1 supplement)

On the Base/Instruct rows of Table 22, we summarize input-side Base-Instruct GCorr under the same convention as the static-spectrum analysis: **13 untied pairs** and **17 tied pairs** (excluding the Gemma-4 family and Gemma-3-1B; see the model-inventory notes above). The summary is in Table 24.

Table 24: Task1 Base-Instruct input-side GCorr summary (13 untied / 17 tied).

Group	n	Metric	mean	range [min, max]
untied	13	E_{euc}	1.000	[0.9998, 1.0000]
untied	13	E_{cos}	0.997	[0.987, 1.000]
tied	17	E_{cos}	0.994	[0.979, 1.000]
tied	17	E_{euc}	0.983	[0.894, 1.000]

Reading guide. Among the 13 untied pairs, E_{euc} is essentially unity ($13/13 \geq 0.999$), indicating that independent input embeddings preserve stable Euclidean token-token geometry across Base-Instruct evolution. Among the 17 tied pairs, E_{cos} remains very high (mean 0.994, same order as untied), but E_{euc} shows clear spread (range 0.106; mean 0.017 lower than untied); the gap is mainly scale drift rather than directional mismatch, and 12/17 pairs satisfy $E_{\text{cos}} > E_{\text{euc}}$. The distance to 1 ($\delta = 1 - \text{mean}$) differs by about $480\times$ between untied and tied E_{euc} (3.6×10^{-5} vs. 1.7×10^{-2}). This supplements Insight 1: tying makes it harder to preserve the stable E-side Euclidean structure of untied input embeddings while keeping cosine directions high; it does not replace the primary Task2 37/37 and static-spectrum evidence.

B.4.2 Task2 cross-model $U-U$ vs. $E-E$ GCorr summary (Insight 2)

On the Task2 cross-model comparisons in Table 22 (110 pairs; excluding Base/Instruct rows), each pair reports both E_{cos} and U_{cos} . Aggregating over all pairs yields **85/110** with $U_{\text{cos}} > E_{\text{cos}}$; mean $U_{\text{cos}} = \mathbf{0.833}$ vs. mean $E_{\text{cos}} = \mathbf{0.484}$; mean gap $\Delta = U_{\text{cos}} - E_{\text{cos}} = \mathbf{+0.349}$. Under Euclidean GCorr, $U_{\text{euc}} > E_{\text{euc}}$ holds for 69/110 (mean $U_{\text{euc}} = 0.855$ vs. $E_{\text{euc}} = 0.777$); Euclidean GCorr is more sensitive to within-family scale choices, so Insight 2 in the main text emphasizes cosine GCorr.

The 25 pairs with $U_{\text{cos}} \leq E_{\text{cos}}$ are listed in Table 25. 24/25 of them do not substantively refute Insight 2: twelve pairs are tied on both sides ($E = U$), so $E_{\text{cos}} \approx U_{\text{cos}}$; twelve DeepSeek same-family checkpoints tie at $E_{\text{cos}} \approx 1$ or differ by only $\lesssim 2 \times 10^{-4}$. Only MiniMax-M2.5-Chat vs.

MiniMax-M2.7 shows a clear $E_{\text{cos}} > U_{\text{cos}}$ exception (0.993 vs. 0.976). The conclusion should therefore be read as a cross-model trend rather than an absolute claim for every pair.

Table 25: The 25 Task2 model pairs that do not satisfy $U_{\text{cos}} > E_{\text{cos}}$ (exceptions among 110 pairs).

Model Pair (A vs. B)	E_{cos}	U_{cos}	Δ	Note
DeepSeek-V3-Base vs. DeepSeek-V3	1.000	1.000	0.000	DeepSeek
DeepSeek-V3.2 vs. DeepSeek-V3.2-Speciale	1.000	1.000	0.000	DeepSeek
DeepSeek-R1 vs. DeepSeek-R1-0528	1.000	1.000	0.000	DeepSeek
DeepSeek-V3.1-Base vs. DeepSeek-V3.1	1.000	1.000	0.000	DeepSeek
DeepSeek-V3-Base vs. DeepSeek-V3.1-Base	1.000	1.000	-0.000	DeepSeek
DeepSeek-V3 vs. DeepSeek-V3.1-Base	1.000	1.000	-0.000	DeepSeek
DeepSeek-V3-Base vs. DeepSeek-V3.1	1.000	1.000	-0.000	DeepSeek
DeepSeek-V3 vs. DeepSeek-V3.1	1.000	1.000	-0.000	DeepSeek
DeepSeek-V3-Base vs. DeepSeek-V3.2-Speciale	1.000	1.000	-0.000	DeepSeek
DeepSeek-V3-Base vs. DeepSeek-V3.2	1.000	1.000	-0.000	DeepSeek
DeepSeek-V3 vs. DeepSeek-V3.2-Speciale	1.000	1.000	-0.000	DeepSeek
DeepSeek-V3 vs. DeepSeek-V3.2	1.000	1.000	-0.000	DeepSeek
MiniMax-M2.5-Chat vs. MiniMax-M2.7	0.993	0.976	-0.017	exception
Qwen3.5-2B-Instruct vs. Qwen3.5-4B-Instruct	0.949	0.949	0.000	tied
Qwen2.5-1.5B-Instruct vs. Qwen2.5-3B-Instruct	0.915	0.915	0.000	tied
Qwen3.5-0.8B-Instruct vs. Qwen3.5-4B-Instruct	0.915	0.915	0.000	tied
Qwen3-0.6B vs. Qwen3-4B	0.884	0.884	0.000	tied
Qwen3-1.7B vs. Qwen3-4B	0.874	0.874	0.000	tied
Qwen3-0.6B vs. Qwen3-1.7B	0.857	0.857	0.000	tied
Qwen2.5-0.5B-Instruct vs. Qwen2.5-1.5B-Instruct	0.851	0.851	0.000	tied
Qwen2.5-0.5B-Instruct vs. Qwen2.5-3B-Instruct	0.848	0.848	0.000	tied
Gemma-3-1B-Instruct vs. Gemma-3-4B-Instruct	0.668	0.668	0.000	tied
Gemma-3-1B-Instruct vs. Gemma-3-12B-Instruct	0.515	0.514	-0.000	tied
Gemma-2-2B-Instruct vs. Gemma-2-27B-Instruct	0.392	0.392	-0.000	tied
Gemma-2-9B-Instruct vs. Gemma-2-27B-Instruct	0.368	0.368	-0.000	tied